Math 135 — Algebra for Honours Mathematics

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Chapter 1

Introduction to Mathematical Language

Math 135 is, for most people, the first course where the *proof* matters more than the answer. Before we can prove anything we need a precise language: what a statement is, how statements combine, and how quantifiers such as "for all" and "there exists" behave. This chapter sets up that vocabulary.

1.1 Statements

Definition 1.1 (Statement)**.** A *statement* (or proposition) is a sentence that is either true or false, but not both.

Example 1.2.

- "7 is prime" is a true statement.
- "$2^{10} < 1000$" is a false statement ($2^{10} = 1024$).
- "Every even integer greater than 2 is a sum of two primes" (Goldbach) is a statement; nobody knows its truth value, but it has one.
- "$x^2 = 4$" is *not* a statement: its truth depends on x. It is an *open sentence*, written $P(x)$.
- "Prove the theorem." is not a statement (it is a command).

An open sentence $P(x)$ becomes a statement once we substitute a value for x or once we *quantify* x (see below). The set of values x is allowed to range over is the *domain* (or universe of discourse).

1.2 Compound statements and truth tables

Given statements A and B we form new statements with the logical connectives:

Symbol	Name	Read as
$\neg A$	negation	"not A"
$A \wedge B$	conjunction	"A and B"
$A \vee B$	disjunction	"A or B" (inclusive)
$A \Rightarrow B$	implication	"if A then B"
$A \Leftrightarrow B$	biconditional	"A if and only if B"

Their meanings are fixed by the truth table:

A	B	$\neg A$	$A \wedge B$	$A \vee B$	$A \Rightarrow B$	$A \Leftrightarrow B$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

The implication is only false in one row

$A \Rightarrow B$ is false *only* when A is true and B is false. In particular, an implication with a false hypothesis is *vacuously true*: "if $1 = 2$ then I am the Pope" is a true statement.

In $A \Rightarrow B$, A is the *hypothesis* and B the *conclusion*. Equivalent English phrasings: "A implies B", "B whenever A", "A is sufficient for B", "B is necessary for A", "A only if B".

1.2.1 Logical equivalence

Definition 1.3. Two compound statements are *logically equivalent*, written $S_1 \equiv S_2$, if they have the same truth value for every assignment of truth values to their component statements. A statement that is true in every row is a *tautology*; one that is false in every row is a *contradiction*.

Proposition 1.4 (Useful equivalences)**.** *For statements A, B, C:*

1. $\neg(\neg A) \equiv A$ *(double negation);*
2. $\neg(A \wedge B) \equiv \neg A \vee \neg B$ *and* $\neg(A \vee B) \equiv \neg A \wedge \neg B$ *(De Morgan's laws);*
3. $A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C)$ *and* $A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$ *(distributive laws);*
4. $A \Rightarrow B \equiv \neg A \vee B$*;*
5. $\neg(A \Rightarrow B) \equiv A \wedge \neg B$*;*
6. $A \Leftrightarrow B \equiv (A \Rightarrow B) \wedge (B \Rightarrow A)$.

Proof of (4). Compare columns:

A	B	$\neg A$	$\neg A \vee B$	$A \Rightarrow B$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

The last two columns agree in every row. The rest are checked the same way. □

Example 1.5. Show $(A \Rightarrow B) \wedge (A \Rightarrow C) \equiv A \Rightarrow (B \wedge C)$ without a table:

$$\begin{aligned}(A \Rightarrow B) \wedge (A \Rightarrow C) &\equiv (\neg A \vee B) \wedge (\neg A \vee C) \\ &\equiv \neg A \vee (B \wedge C) && \text{(distributive law)} \\ &\equiv A \Rightarrow (B \wedge C).\end{aligned}$$

1.3 Converse and contrapositive

Definition 1.6. For the implication $A \Rightarrow B$:

- the *converse* is $B \Rightarrow A$;
- the *contrapositive* is $\neg B \Rightarrow \neg A$;
- the *inverse* is $\neg A \Rightarrow \neg B$.

Theorem 1.7. *An implication is logically equivalent to its contrapositive:* $A \Rightarrow B \equiv \neg B \Rightarrow \neg A$.

Proof. $\neg B \Rightarrow \neg A \equiv \neg(\neg B) \vee \neg A \equiv B \vee \neg A \equiv \neg A \vee B \equiv A \Rightarrow B$. □

Converse is not equivalent

The converse is *not* equivalent to the original. "If $n = 4$ then n is even" is true; its converse "if n is even then $n = 4$" is false (take $n = 2$). The inverse is the contrapositive of the converse, so the inverse is equivalent to the converse.

1.4 Quantifiers

Definition 1.8. Let $P(x)$ be an open sentence with domain S.

- The *universal* statement $\forall x \in S,\ P(x)$ is true if $P(x)$ is true for every $x \in S$.
- The *existential* statement $\exists x \in S,\ P(x)$ is true if $P(x)$ is true for at least one $x \in S$.

Example 1.9.

- $\forall x \in \mathbb{R},\ x^2 \geq 0$ is true.
- $\exists x \in \mathbb{R},\ x^2 = -1$ is false; but $\exists x \in \mathbb{C},\ x^2 = -1$ is true. The domain matters.
- $\forall n \in \mathbb{N},\ n^2 + n + 41$ is prime is false: at $n = 40$ we get $40^2 + 40 + 41 = 1681 = 41^2$.

1.4.1 Nested quantifiers

The order of quantifiers of different types matters.

Example 1.10.

- $\forall x \in \mathbb{R},\ \exists y \in \mathbb{R},\ x + y = 0$ is true: given x, choose $y = -x$. Here y may depend on x.
- $\exists y \in \mathbb{R},\ \forall x \in \mathbb{R},\ x + y = 0$ is false: a single y would have to satisfy $1 + y = 0$ and $2 + y = 0$ at once.

Quantifiers of the *same* type can be swapped: $\forall x\, \forall y$ is the same as $\forall y\, \forall x$.

1.4.2 Negating quantified statements

Theorem 1.11.

$$\neg(\forall x \in S,\ P(x)) \equiv \exists x \in S,\ \neg P(x), \qquad \neg(\exists x \in S,\ P(x)) \equiv \forall x \in S,\ \neg P(x).$$

To negate a long statement, move the negation from left to right, flipping each quantifier and finally negating the innermost open sentence (using De Morgan's laws and $\neg(A \Rightarrow B) \equiv A \wedge \neg B$).

Example 1.12. Negate "for every $\varepsilon > 0$ there is a $\delta > 0$ such that for all $x \in \mathbb{R}$, if $|x| < \delta$ then $|f(x)| < \varepsilon$".

$$\begin{aligned}&\neg(\forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x \in \mathbb{R},\ (|x| < \delta \Rightarrow |f(x)| < \varepsilon))\\ &\equiv \exists \varepsilon > 0,\ \forall \delta > 0,\ \exists x \in \mathbb{R},\ \neg(|x| < \delta \Rightarrow |f(x)| < \varepsilon)\\ &\equiv \exists \varepsilon > 0,\ \forall \delta > 0,\ \exists x \in \mathbb{R},\ (|x| < \delta \wedge |f(x)| \geq \varepsilon).\end{aligned}$$

Example 1.13. Negate "every integer greater than 1 has a prime divisor": "there exists an integer greater than 1 that has no prime divisor". Note the negation is *not* "no integer greater than 1 has a prime divisor".

1.5 Reading and writing mathematics

A theorem typically has the shape "for all x in S, if $H(x)$ then $C(x)$", even when the quantifier is hidden: "if n is odd then n^2 is odd" really means $\forall n \in \mathbb{Z},\ (n \text{ odd} \Rightarrow n^2 \text{ odd})$. Before proving something, identify:

1. the quantifiers and the domain;
2. the hypothesis (what we may assume);
3. the conclusion (what we must show).

Common notation: $\mathbb{N} = \{1, 2, 3, \dots\}$ (in Math 135 the natural numbers start at 1), $\mathbb{Z}$ integers, $\mathbb{Q}$ rationals, $\mathbb{R}$ reals, $\mathbb{C}$ complex numbers. "Such that" is often abbreviated "s.t."; $\therefore$ means "therefore".

1.6 Common pitfalls

- Treating "or" as exclusive. In mathematics $A \vee B$ is true when both are true.
- Thinking a false hypothesis makes an implication false. It makes it (vacuously) true.
- Proving the converse instead of the statement.
- Negating $\forall x,\ P(x)$ as $\forall x,\ \neg P(x)$. The correct negation needs only one counterexample.
- Swapping $\forall\exists$ to $\exists\forall$, which usually makes a statement much stronger.

1.7 Summary

- A statement is true or false; an open sentence becomes a statement when its variables are assigned or quantified.
- Connectives $\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow$ are defined by truth tables; $A \Rightarrow B$ is false only when A is true and B false.
- $A \Rightarrow B \equiv \neg A \vee B \equiv \neg B \Rightarrow \neg A$ (contrapositive); the converse $B \Rightarrow A$ is a different statement.
- Negation flips quantifiers: $\neg\forall \equiv \exists\neg$ and $\neg\exists \equiv \forall\neg$; the order of mixed quantifiers matters.

Chapter 2

Proof Techniques

A proof is a finite chain of statements, each of which is an assumption, a previously proved result, or follows logically from earlier lines, ending in the statement to be proved. This chapter collects the standard proof templates used throughout the course.

2.1 Proving universal and existential statements

- To prove $\forall x \in S,\ P(x)$: let x be an *arbitrary* element of S (do not pick a specific one) and show $P(x)$.
- To prove $\exists x \in S,\ P(x)$: exhibit (construct) a specific $x \in S$ and verify $P(x)$. You do not need to explain how you found it.
- To *disprove* $\forall x \in S,\ P(x)$: give a single *counterexample* $x \in S$ with $P(x)$ false.
- To disprove $\exists x \in S,\ P(x)$: prove $\forall x \in S,\ \neg P(x)$.

Example 2.1. Prove there is an integer n such that $n^2 - 5n + 6 = 0$. *Proof.* Take $n = 2$: $4 - 10 + 6 = 0$. □

Example 2.2 (Counterexample)**.** Disprove: "for all real a, b, if $a^2 = b^2$ then $a = b$". Take $a = 1$, $b = -1$: then $a^2 = b^2 = 1$ but $a \neq b$.

2.2 Direct proof

To prove $A \Rightarrow B$ directly, assume A and deduce B. Many proofs in this course start from definitions, so it pays to state them exactly.

Definition 2.3. An integer n is *even* if $n = 2k$ for some $k \in \mathbb{Z}$, and *odd* if $n = 2k + 1$ for some $k \in \mathbb{Z}$. For integers a, b, we say a *divides* b, written $a \mid b$, if there is $k \in \mathbb{Z}$ with $b = ka$.

Proposition 2.4. *If n is an odd integer then n^2 is odd.*

Proof. Let n be odd, so $n = 2k + 1$ for some $k \in \mathbb{Z}$. Then $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$. Since $2k^2 + 2k \in \mathbb{Z}$, n^2 is odd. □

Proposition 2.5. *For all $a, b, c \in \mathbb{Z}$, if $a \mid b$ and $a \mid c$ then $a \mid (b + c)$.*

Proof. Since $a \mid b$ and $a \mid c$ there are integers k, ℓ with $b = ka$ and $c = \ell a$. Then $b + c = (k + \ell)a$ with $k + \ell \in \mathbb{Z}$, so $a \mid (b + c)$. □

Working backwards is not a proof

When solving, it is fine to start from the conclusion and work backwards on scratch paper, but the written proof must go from hypothesis to conclusion with each step justified. In particular, "$B \Rightarrow \mathrm{T}$" does not prove B: from $-1 = 1$ one gets $1 = 1$ by squaring.

Example 2.6 (AM–GM)**.** Prove that for all real $x, y \geq 0$, $\sqrt{xy} \leq \frac{x+y}{2}$.

Scratch work. $\sqrt{xy} \leq \frac{x+y}{2} \iff 2\sqrt{xy} \leq x + y \iff 0 \leq x - 2\sqrt{xy} + y = (\sqrt{x} - \sqrt{y})^2$.

Proof. Since $\sqrt{x}, \sqrt{y}$ are real, $(\sqrt{x} - \sqrt{y})^2 \geq 0$, i.e. $x - 2\sqrt{xy} + y \geq 0$. Rearranging, $\sqrt{xy} \leq \frac{x+y}{2}$. □

2.3 Proof by cases

If $A_1 \vee A_2 \vee \cdots \vee A_k$ is known to hold (the cases are exhaustive) and each $A_i \Rightarrow B$, then B holds.

Proposition 2.7. *For every integer n, $n^2 + n$ is even.*

Proof. Case 1: n even, $n = 2k$. Then $n^2 + n = 4k^2 + 2k = 2(2k^2 + k)$, even. *Case 2: n odd,* $n = 2k + 1$. Then $n^2 + n = (2k + 1)(2k + 2) = 2(2k + 1)(k + 1)$, even. Every integer is even or odd, so the result holds. □

2.4 Proof by contrapositive

Since $A \Rightarrow B \equiv \neg B \Rightarrow \neg A$, we may instead assume $\neg B$ and derive $\neg A$. This helps when $\neg B$ gives something concrete to work with.

Proposition 2.8. *Let $n \in \mathbb{Z}$. If n^2 is even, then n is even.*

Proof. We prove the contrapositive: if n is odd then n^2 is odd. This was proved above. □

Proposition 2.9. *For $x \in \mathbb{R}$, if $x^3 + 5x < 6$ then $x < 1$.*

Proof. Contrapositive: suppose $x \geq 1$. Then $x^3 \geq 1$ and $5x \geq 5$, so $x^3 + 5x \geq 6$. □

2.5 Proof by contradiction

To prove B, assume $\neg B$ and derive a contradiction (a statement of the form $C \wedge \neg C$). Then $\neg B$ must be false, so B is true.

Theorem 2.10. *$\sqrt{2}$ is irrational.*

Proof. Suppose, for a contradiction, that $\sqrt{2} = p/q$ with $p, q \in \mathbb{Z}$, $q > 0$, and the fraction in lowest terms (so p and q are not both even). Then $p^2 = 2q^2$, so p^2 is even, hence p is even by the proposition above. Write $p = 2k$; then $4k^2 = 2q^2$, so $q^2 = 2k^2$ is even and q is even. Both p and q are even, a contradiction. □

Theorem 2.11 (Euclid)**.** *There are infinitely many primes.*

Proof. Suppose there are finitely many, $p_1, \dots, p_k$. Let $N = p_1 p_2 \cdots p_k + 1 > 1$. N has some prime divisor (every integer > 1 does, Chapter 5), which must be some p_i. Then $p_i \mid N$ and $p_i \mid p_1 \cdots p_k$, so $p_i \mid N - p_1 \cdots p_k = 1$, which is impossible. □

Contradiction versus contrapositive: to prove $A \Rightarrow B$ by contradiction we assume A and $\neg B$ and reach *any* contradiction; the contrapositive is the special case where we derive $\neg A$.

2.6 If and only if

To prove $A \Leftrightarrow B$ prove both directions, $A \Rightarrow B$ and $B \Rightarrow A$ (each by any method). Alternatively build a chain of equivalences $A \iff C_1 \iff \cdots \iff B$, being careful that every step really is reversible.

Proposition 2.12. *Let $n \in \mathbb{Z}$. Then n is even if and only if n^2 is even.*

Proof. ($\Rightarrow$) If $n = 2k$ then $n^2 = 2(2k^2)$ is even. ($\Leftarrow$) Proved above by contrapositive. □

Example 2.13. Prove that for $x \in \mathbb{R}$, $x^2 - 3x + 2 < 0 \iff 1 < x < 2$. We have $x^2 - 3x + 2 = (x-1)(x-2)$. A product of two reals is negative if and only if the factors have opposite signs. $x - 1 < 0 < x - 2$ is impossible (it would need $x < 1$ and $x > 2$), so the product is negative iff $x - 1 > 0 > x - 2$, i.e. iff $1 < x < 2$.

2.7 Uniqueness

To prove "there is a unique $x \in S$ with $P(x)$" (written $\exists!\, x$):

1. *Existence*: find some x with $P(x)$.
2. *Uniqueness*: assume $P(x)$ and $P(y)$ and show $x = y$; or assume $x \neq y$ both satisfy P and derive a contradiction.

Proposition 2.14. *For every $a \in \mathbb{R}$ and $b \neq 0$, the equation $bx = a$ has a unique real solution.*

Proof. Existence: $x = a/b$ works since $b(a/b) = a$. *Uniqueness:* if $bx = a$ and $by = a$ then $b(x - y) = 0$; since $b \neq 0$, $x = y$. □

2.8 Common pitfalls

- Proving a universal statement by checking examples. Examples can only disprove.
- Using the same letter for two different unknowns: if m and n are both odd, write $m = 2k + 1$ and $n = 2\ell + 1$, not both with k.
- Dividing by something that might be zero.
- In a contradiction proof, forgetting to say what the contradiction is.

2.9 Summary

- $\forall$: take an arbitrary element. $\exists$: construct a witness. To disprove $\forall$: one counterexample.
- Direct proof: assume the hypothesis, reach the conclusion, using definitions.
- Contrapositive: prove $\neg B \Rightarrow \neg A$. Contradiction: assume the statement is false and derive an impossibility.
- $A \Leftrightarrow B$: prove both directions. Uniqueness: existence plus "any two are equal".

Chapter 3

Sets

3.1 Basic notions

Definition 3.1. A *set* is a collection of objects, called its *elements*. We write $x \in S$ if x is an element of S and $x \notin S$ otherwise. The *empty set* $\emptyset$ has no elements. A set with finitely many elements is finite; its number of elements is its *cardinality*, $|S|$.

Sets are described either by listing, $\{1, 2, 3\}$, or with *set-builder notation* $\{x \in U : P(x)\}$, "the set of x in U such that $P(x)$". Order and repetition do not matter: $\{1, 2, 2, 3\} = \{3, 1, 2\}$.

Example 3.2.

- $\{n \in \mathbb{Z} : n^2 < 10\} = \{-3, -2, -1, 0, 1, 2, 3\}$.
- $\{2k + 1 : k \in \mathbb{Z}\}$ is the set of odd integers.
- $\{\emptyset\} \neq \emptyset$: the former has one element.
- $|\{1, \{1, 2\}, \emptyset\}| = 3$.

3.2 Set operations

Let A, B be subsets of a universal set U.

Definition 3.3.

- Union: $A \cup B = \{x : x \in A \vee x \in B\}$.
- Intersection: $A \cap B = \{x : x \in A \wedge x \in B\}$.
- Difference: $A - B = A \setminus B = \{x : x \in A \wedge x \notin B\}$.
- Complement: $\bar{A} = A^c = \{x \in U : x \notin A\}$.

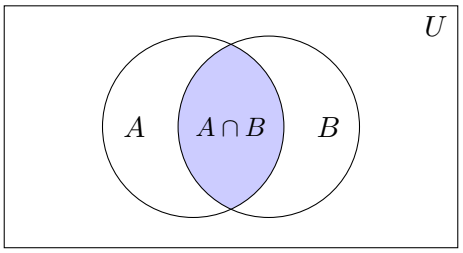
A and B are *disjoint* if $A \cap B = \emptyset$.

The definitions translate set operations into the logical connectives of Chapter 1: $\cup$ is $\vee$, $\cap$ is $\wedge$, complement is $\neg$. Consequently set identities mirror logical equivalences.

Example 3.4. Let $U = \{1, \ldots, 10\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{2, 4, 6, 8, 10\}$. Then $A \cup B = \{1, 2, 3, 4, 5, 6, 8, 10\}$, $A \cap B = \{2, 4\}$, $A - B = \{1, 3, 5\}$, $\bar{B} = \{1, 3, 5, 7, 9\}$.

Theorem 3.5 (De Morgan's laws for sets)**.** *$\overline{A \cup B} = \bar{A} \cap \bar{B}$ and $\overline{A \cap B} = \bar{A} \cup \bar{B}$.*

Venn diagrams are useful for guessing whether an identity is true, but are not accepted as a proof.



3.3 Subsets and set equality

Definition 3.6. A is a *subset* of B, $A \subseteq B$, if every element of A is an element of B: $\forall x,\ x \in A \Rightarrow x \in B$. A is a *proper subset*, $A \subsetneq B$, if additionally $A \neq B$. Two sets are *equal* if $A \subseteq B$ and $B \subseteq A$.

Templates

- To show $A \subseteq B$: "Let $x \in A$. ... Therefore $x \in B$."
- To show $A = B$: show $A \subseteq B$ and $B \subseteq A$ (two "element chasing" arguments), or a chain of equivalences $x \in A \iff \cdots \iff x \in B$.
- To show $A \not\subseteq B$: find one $x \in A$ with $x \notin B$.

Note $\emptyset \subseteq A$ for every set A (vacuously), and $\in$ is not the same as $\subseteq$: $1 \in \{1, 2\}$ but $\{1\} \subseteq \{1, 2\}$.

Example 3.7. Show $\{n \in \mathbb{Z} : 6 \mid n\} \subseteq \{n \in \mathbb{Z} : 2 \mid n\}$. Let n be in the first set, so $n = 6k$ for some $k \in \mathbb{Z}$. Then $n = 2(3k)$, so $2 \mid n$ and n is in the second set. The reverse inclusion fails: 2 is in the second set but not the first.

Proposition 3.8. *For sets A, B, C: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.*

Proof. ($\subseteq$) Let $x \in A \cap (B \cup C)$. Then $x \in A$ and ($x \in B$ or $x \in C$). If $x \in B$ then $x \in A \cap B$; if $x \in C$ then $x \in A \cap C$. Either way $x \in (A \cap B) \cup (A \cap C)$.

($\supseteq$) Let $x \in (A \cap B) \cup (A \cap C)$. If $x \in A \cap B$, then $x \in A$ and $x \in B \subseteq B \cup C$. If $x \in A \cap C$, then $x \in A$ and $x \in C \subseteq B \cup C$. In both cases $x \in A \cap (B \cup C)$. □

Proposition 3.9. *$A \subseteq B$ if and only if $A \cap B = A$.*

Proof. ($\Rightarrow$) Assume $A \subseteq B$. Always $A \cap B \subseteq A$. Conversely if $x \in A$ then $x \in B$ by assumption, so $x \in A \cap B$. Hence $A \cap B = A$. ($\Leftarrow$) Assume $A \cap B = A$ and let $x \in A$. Then $x \in A \cap B$, so $x \in B$. □

Example 3.10 (A non-obvious equality)**.** Let S be the set of integers leaving remainder 1 or 3 when divided by 4, and T the odd integers. Then $S = T$: if $n = 4k+1$ or $4k+3$ then $n = 2(2k)+1$ or $2(2k+1)+1$, odd. Conversely if $n = 2m+1$, then m is even ($m = 2k$, $n = 4k+1$) or odd ($m = 2k+1$, $n = 4k+3$).

Example 3.11 (Disproving an identity)**.** Is $A - (B - C) = (A - B) - C$? Try $A = B = C = \{1\}$: $A - (B - C) = \{1\} - \emptyset = \{1\}$, while $(A - B) - C = \emptyset - \{1\} = \emptyset$. So the identity is false.

3.4 Cartesian products and power sets

Definition 3.12. The *Cartesian product* of A and B is the set of ordered pairs

$$A \times B = \{(a, b) : a \in A,\ b \in B\}.$$

Ordered pairs satisfy $(a, b) = (c, d) \iff a = c$ and $b = d$. If A and B are finite then $|A \times B| = |A|\,|B|$.

Example 3.13. $\{1, 2\} \times \{x, y, z\} = \{(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)\}$, which has $2 \cdot 3 = 6$ elements. Note $A \times B \neq B \times A$ in general. $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$ is the plane.

Proposition 3.14. $A \times (B \cap C) = (A \times B) \cap (A \times C)$.

Proof. For any pair (x, y):

$$\begin{aligned}(x, y) \in A \times (B \cap C) &\iff x \in A \wedge (y \in B \wedge y \in C) \\ &\iff (x \in A \wedge y \in B) \wedge (x \in A \wedge y \in C) \\ &\iff (x, y) \in (A \times B) \cap (A \times C).\end{aligned}$$

Every element of either side is an ordered pair, so the sets are equal. □

Definition 3.15. The *power set* $\mathcal{P}(A)$ is the set of all subsets of A. If $|A| = n$ then $|\mathcal{P}(A)| = 2^n$ (each element is either in or out of a subset).

Example 3.16. $\mathcal{P}(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$.

3.5 Common pitfalls

- Confusing $\in$ with $\subseteq$, or $\emptyset$ with $\{\emptyset\}$.
- Proving only one inclusion when asked for equality.
- Starting an inclusion proof with a specific element instead of an arbitrary one.
- Using a Venn diagram as the proof.

3.6 Summary

- Union, intersection, complement correspond to $\vee$, $\wedge$, $\neg$; set identities follow from logical equivalences.
- $A \subseteq B$: let $x \in A$ and show $x \in B$. $A = B$: show both inclusions.
- $A \times B$ is the set of ordered pairs; $|A \times B| = |A|\,|B|$; $|\mathcal{P}(A)| = 2^{|A|}$.

Chapter 4

Induction

Induction is the main tool for proving statements of the form "for all $n \in \mathbb{N}$, $P(n)$". Math 135 also uses it to justify recursive definitions and to prove the binomial theorem.

4.1 Sigma and product notation

$\sum_{i=1}^{n} a_i = a_1 + a_2 + \cdots + a_n$ and $\prod_{i=1}^{n} a_i = a_1 a_2 \cdots a_n$. Standard manipulations: $\sum(a_i + b_i) = \sum a_i + \sum b_i$, $\sum c a_i = c \sum a_i$, and the index can be shifted: $\sum_{i=1}^{n} a_i = \sum_{j=0}^{n-1} a_{j+1}$. An empty sum is 0 and an empty product is 1.

4.2 The principle of mathematical induction

Theorem 4.1 (Principle of Mathematical Induction (POMI))**.** *Let $P(n)$ be an open sentence with domain $\mathbb{N}$. If*

1. *(base case) $P(1)$ is true, and*
2. *(inductive step) for every $k \in \mathbb{N}$, $P(k) \Rightarrow P(k+1)$,*

then $P(n)$ is true for all $n \in \mathbb{N}$.

The base case can start at any integer b, giving $P(n)$ for all $n \geq b$. In the inductive step, the assumption $P(k)$ is called the *inductive hypothesis* (IH).

Template

Base case. Verify $P(b)$ explicitly (compute both sides).
Inductive hypothesis. Assume $P(k)$ holds for some $k \geq b$.
Inductive step. Show $P(k+1)$, pointing out where the IH is used.
Conclusion. By POMI, $P(n)$ holds for all $n \geq b$.

Example 4.2. Prove $\sum_{i=1}^{n} i^2 = \dfrac{n(n+1)(2n+1)}{6}$ for all $n \in \mathbb{N}$.

Base case $n = 1$: LHS $= 1$, RHS $= \frac{1 \cdot 2 \cdot 3}{6} = 1$.

Inductive step. Assume $\sum_{i=1}^{k} i^2 = \frac{k(k+1)(2k+1)}{6}$ for some $k \in \mathbb{N}$. Then

$$\begin{aligned}\sum_{i=1}^{k+1} i^2 &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \qquad \text{(IH)}\\ &= \frac{(k+1)\big[k(2k+1)+6(k+1)\big]}{6} = \frac{(k+1)(2k^2+7k+6)}{6}\\ &= \frac{(k+1)(k+2)(2k+3)}{6},\end{aligned}$$

which is the formula with $n = k+1$. By POMI the formula holds for all n.

Example 4.3 (Divisibility)**.** Prove $3 \mid 4^n - 1$ for all $n \geq 0$. Base: $4^0 - 1 = 0 = 3 \cdot 0$. Step: assume $4^k - 1 = 3m$. Then $4^{k+1} - 1 = 4 \cdot 4^k - 1 = 4(3m+1) - 1 = 12m + 3 = 3(4m+1)$.

Example 4.4 (Inequality with a later base case)**.** Prove $2^n > n^2$ for all $n \geq 5$. Base: $2^5 = 32 > 25$. Step: assume $2^k > k^2$ with $k \geq 5$. Then $2^{k+1} = 2 \cdot 2^k > 2k^2 = k^2 + k^2$. Since $k \geq 5$, $k^2 \geq 5k = 2k + 3k > 2k + 1$, so $2^{k+1} > k^2 + 2k + 1 = (k+1)^2$. (The claim fails for $n = 4$: $16 = 16$.)

4.3 Strong induction

Theorem 4.5 (Principle of Strong Induction (POSI))**.** *Let $P(n)$ be an open sentence. If $P(b), P(b+1), \ldots, P(b+r)$ are true (several base cases) and for every $k \geq b + r$,*

$$P(b) \wedge P(b+1) \wedge \cdots \wedge P(k) \Rightarrow P(k+1),$$

then $P(n)$ holds for all $n \geq b$.

Strong induction is needed when $P(k+1)$ depends on some earlier case other than $P(k)$, such as in recurrences of depth 2 or in factorization arguments. The number of base cases must be enough that the inductive step only ever refers to cases already covered.

Example 4.6 (Recurrence)**.** Define $x_1 = 3$, $x_2 = 5$, and $x_n = 3x_{n-1} - 2x_{n-2}$ for $n \geq 3$. Prove $x_n = 2^n + 1$ for all $n \in \mathbb{N}$.

Base cases: $x_1 = 3 = 2^1 + 1$ and $x_2 = 5 = 2^2 + 1$.

Inductive step: let $k \geq 2$ and assume $x_i = 2^i + 1$ for all $1 \leq i \leq k$. Then

$$\begin{aligned}x_{k+1} = 3x_k - 2x_{k-1} &= 3(2^k + 1) - 2(2^{k-1} + 1)\\ &= 3 \cdot 2^k - 2^k + 3 - 2 = 2 \cdot 2^k + 1 = 2^{k+1} + 1.\end{aligned}$$

Both k and $k-1$ are in $\{1, \ldots, k\}$ because $k \geq 2$; this is why two base cases are needed. Check: $x_3 = 15 - 6 = 9 = 2^3 + 1$.

Example 4.7 (Fibonacci bound)**.** Let $f_1 = f_2 = 1$ and $f_n = f_{n-1} + f_{n-2}$. Prove $f_n < 2^n$. Base: $f_1 = 1 < 2$, $f_2 = 1 < 4$. Step ($k \geq 2$): $f_{k+1} = f_k + f_{k-1} < 2^k + 2^{k-1} < 2^k + 2^k = 2^{k+1}$.

Proposition 4.8. *Every integer $n \geq 2$ can be written as a product of primes.*

Proof. Strong induction on n. Base: 2 is prime. Suppose every integer m with $2 \leq m \leq k$ is a product of primes, and consider $k+1$. If $k+1$ is prime we are done. Otherwise $k+1 = ab$ with $2 \leq a, b \leq k$. By the IH both a and b are products of primes, so $k+1 = ab$ is too. □

4.4 The well-ordering principle

Theorem 4.9 (Well-Ordering Principle (WOP))**.** *Every non-empty subset of* $\mathbb{N}$ *has a smallest element.*

WOP is equivalent to induction. It gives an alternative "minimal counterexample" style of proof: if $P(n)$ failed for some n, take the *smallest* such n and derive a contradiction, usually by producing a smaller counterexample.

Example 4.10. Reprove that every $n \geq 2$ is a product of primes with WOP. Suppose not; let S be the non-empty set of counterexamples and $n = \min S$. Then n is not prime, so $n = ab$ with $1 < a, b < n$. By minimality, $a, b \notin S$, so both are products of primes, and so is n. Contradiction.

4.5 Recursively defined sequences

A sequence is defined *recursively* by giving initial terms and a rule expressing x_n in terms of earlier terms. A common task is to guess a closed form by computing a few terms and then prove it by (strong) induction.

Example 4.11. $a_1 = 1$, $a_n = 2a_{n-1} + 1$ for $n \geq 2$. Terms: $1, 3, 7, 15, 31$, suggesting $a_n = 2^n - 1$. Induction: $a_1 = 1 = 2^1 - 1$, and if $a_k = 2^k - 1$ then $a_{k+1} = 2(2^k - 1) + 1 = 2^{k+1} - 1$.

4.6 The binomial theorem

Definition 4.12. For integers $0 \leq k \leq n$, $\dbinom{n}{k} = \dfrac{n!}{k!(n-k)!}$, the number of k-element subsets of an n-element set. By convention $\binom{n}{k} = 0$ if $k < 0$ or $k > n$.

Lemma 4.13 (Pascal's identity)**.** *For* $n \geq 1$ *and* $1 \leq k \leq n$*:* $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$.

Proof. For $1 \leq k \leq n-1$,

$$\binom{n-1}{k-1} + \binom{n-1}{k} = \frac{(n-1)!}{(k-1)!(n-k)!} + \frac{(n-1)!}{k!(n-1-k)!} = \frac{(n-1)!\,[k + (n-k)]}{k!(n-k)!} = \binom{n}{k}.$$

For $k = n$ both sides equal 1. □

Theorem 4.14 (Binomial theorem)**.** *For all* $n \geq 0$ *and* a, b *(real or complex),*

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k.$$

Proof. Induction on n. For $n = 0$ both sides are 1. Assume the result for n. Then

$$\begin{aligned}
(a+b)^{n+1} &= (a+b) \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k = \sum_{k=0}^{n} \binom{n}{k} a^{n+1-k} b^k + \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^{k+1} \\
&= \sum_{k=0}^{n+1} \binom{n}{k} a^{n+1-k} b^k + \sum_{k=0}^{n+1} \binom{n}{k-1} a^{n+1-k} b^k \qquad \text{(shift index; added terms are 0)} \\
&= \sum_{k=0}^{n+1} \left[\binom{n}{k} + \binom{n}{k-1} \right] a^{n+1-k} b^k = \sum_{k=0}^{n+1} \binom{n+1}{k} a^{n+1-k} b^k
\end{aligned}$$

by Pascal's identity (the $k = 0$ term is $\binom{n}{0} = 1 = \binom{n+1}{0}$). □

Example 4.15. The coefficient of x^5 in $(2x-3)^8$: the general term is $\binom{8}{k}(2x)^{8-k}(-3)^k$; we need $8-k=5$, so $k=3$, giving $\binom{8}{3}2^5(-3)^3 = 56\cdot 32\cdot(-27) = -48384$.

Example 4.16. Setting $a=b=1$: $\sum_{k=0}^{n}\binom{n}{k} = 2^n$. Setting $a=1$, $b=-1$ (for $n\geq 1$): $\sum_{k=0}^{n}(-1)^k\binom{n}{k} = 0$.

4.7 Common pitfalls

- Skipping or fudging the base case. The "proof" that all horses are the same colour fails because the step $P(1)\Rightarrow P(2)$ is invalid.
- Assuming what you need to prove ("assume $P(n)$ for all n").
- Using too few base cases in strong induction.
- Not stating where the inductive hypothesis is used.

4.8 Summary

- POMI: base case plus $P(k)\Rightarrow P(k+1)$. POSI: enough base cases plus $P(b)\wedge\cdots\wedge P(k)\Rightarrow P(k+1)$.
- WOP (every non-empty subset of $\mathbb{N}$ has a least element) is equivalent and supports minimal-counterexample proofs.
- Recurrences: guess a closed form, prove it by strong induction.
- Binomial theorem $(a+b)^n = \sum\binom{n}{k}a^{n-k}b^k$, proved via Pascal's identity.

Chapter 5

Divisibility and the Division Algorithm

From here on the course becomes elementary number theory: the integers, divisibility, and eventually modular arithmetic and cryptography.

5.1 Divisibility

Definition 5.1. For $a, b \in \mathbb{Z}$, a *divides* b, written $a \mid b$, if there exists $k \in \mathbb{Z}$ such that $b = ka$. We also say a is a *divisor* (factor) of b and b is a *multiple* of a. We write $a \nmid b$ otherwise.

Example 5.2. $3 \mid 12$, $-3 \mid 12$, $5 \nmid 12$, $7 \mid 0$ (since $0 = 0 \cdot 7$), and $0 \mid b$ only for $b = 0$. Every integer divides 0, and ± 1 divide everything.

Divides is a statement, not a number

$a \mid b$ is a statement (true or false); a/b is a number. Never write things like "$3 \mid 12 = 4$".

Proposition 5.3 (Transitivity of Divisibility, TD)**.** *For $a, b, c \in \mathbb{Z}$: if $a \mid b$ and $b \mid c$ then $a \mid c$.*

Proof. Write $b = ka$ and $c = \ell b$. Then $c = (\ell k)a$ with $\ell k \in \mathbb{Z}$. □

Proposition 5.4 (Divisibility of Integer Combinations, DIC)**.** *For $a, b, c \in \mathbb{Z}$: if $a \mid b$ and $a \mid c$, then $a \mid (bx + cy)$ for all $x, y \in \mathbb{Z}$.*

Proof. Write $b = ka$ and $c = \ell a$. Then $bx + cy = (kx + \ell y)a$, and $kx + \ell y \in \mathbb{Z}$. □

DIC is used constantly. A typical use: if a divides two numbers, it divides their sum and difference.

Example 5.5. Let $n \in \mathbb{Z}$. Prove that if $d \mid (5n + 3)$ and $d \mid (3n + 2)$ then $d = \pm 1$. By DIC with $x = 3$, $y = -5$: $d \mid 3(5n + 3) - 5(3n + 2) = 15n + 9 - 15n - 10 = -1$, and the only divisors of -1 are ± 1.

Proposition 5.6 (Bounds by Divisibility, BBD)**.** *If $a \mid b$ and $b \neq 0$, then $|a| \leq |b|$.*

Proof. $b = ka$ with $k \neq 0$ (since $b \neq 0$), so $|k| \geq 1$ and $|b| = |k|\,|a| \geq |a|$. □

Corollary 5.7. *If $a \mid b$ and $b \mid a$ then $a = \pm b$.*

5.2 The division algorithm

Theorem 5.8 (Division Algorithm)**.** *For all $a \in \mathbb{Z}$ and $b \in \mathbb{N}$, there exist* unique *integers q (quotient) and r (remainder) such that*

$$a = qb + r, \qquad 0 \le r < b.$$

Proof. Existence. Consider $S = \{a - qb : q \in \mathbb{Z},\ a - qb \ge 0\}$. S is non-empty: take $q = -|a|$, then $a - qb = a + |a|\, b \ge a + |a| \ge 0$. By the well-ordering principle (applied to the non-negative integers, which works the same way as for $\mathbb{N}$) S has a smallest element $r = a - qb \ge 0$. If $r \ge b$ then $r - b = a - (q+1)b \ge 0$ would be a smaller element of S, contradiction. So $0 \le r < b$.

Uniqueness. Suppose $a = q_1 b + r_1 = q_2 b + r_2$ with $0 \le r_1, r_2 < b$. Then $(q_1 - q_2)b = r_2 - r_1$, so $b \mid r_2 - r_1$. But $-b < r_2 - r_1 < b$, and the only multiple of b strictly between $-b$ and b is 0. So $r_1 = r_2$, and then $q_1 = q_2$. □

Example 5.9.

- $a = 47$, $b = 6$: $47 = 7 \cdot 6 + 5$, so $q = 7$, $r = 5$.
- $a = -17$, $b = 5$: $-17 = (-4) \cdot 5 + 3$, so $q = -4$, $r = 3$. (Not $q = -3$, $r = -2$: the remainder must be non-negative.)
- $a = 30$, $b = 5$: $q = 6$, $r = 0$. In general $b \mid a \iff r = 0$.

Example 5.10 (Using the division algorithm for cases)**.** Prove that the square of any integer has the form $4k$ or $4k + 1$. By the division algorithm with $b = 2$, $n = 2m$ or $n = 2m + 1$. Then $n^2 = 4m^2$ or $n^2 = 4(m^2 + m) + 1$. Consequently no integer of the form $4k + 3$ (such as 2023) is a perfect square, and a sum of two squares is never $\equiv 3$ on division by 4.

Example 5.11. Prove that $n^3 - n$ is divisible by 3 for every integer n. Write $n = 3m + r$ with $r \in \{0, 1, 2\}$. Since $n^3 - n = (n-1)n(n+1)$, one of the three consecutive integers $n - 1, n, n + 1$ is a multiple of 3: if $r = 0$ it is n, if $r = 1$ it is $n - 1 = 3m$, if $r = 2$ it is $n + 1 = 3(m + 1)$. By DIC (or directly) $3 \mid n^3 - n$.

5.3 Primes

Definition 5.12. An integer $p > 1$ is *prime* if its only positive divisors are 1 and p. An integer $n > 1$ that is not prime is *composite*: $n = ab$ with $1 < a, b < n$.

Proposition 5.13. *Every integer $n > 1$ has a prime divisor. Moreover, if n is composite it has a prime divisor $p \le \sqrt{n}$.*

Proof. Let p be the smallest divisor of n with $p > 1$ (exists by WOP; n itself is one). If p were composite, $p = ab$ with $1 < a < p$, and $a \mid p \mid n$ would contradict minimality (by TD). So p is prime. If $n = ab$ is composite with $1 < a \le b$, then $a^2 \le ab = n$, so $a \le \sqrt{n}$, and any prime divisor of a is at most $\sqrt{n}$ and divides n. □

Example 5.14 (Primality test)**.** Is 221 prime? $\sqrt{221} < 15$, so test primes $2, 3, 5, 7, 11, 13$: $221 = 13 \cdot 17$. Composite. Is 223 prime? It is odd; digit sum 7, so $3 \nmid 223$; it does not end in 0 or 5; $223 = 7 \cdot 31 + 6$, $223 = 11 \cdot 20 + 3$, $223 = 13 \cdot 17 + 2$. No prime up to $\sqrt{223} \approx 14.9$ divides it, so 223 is prime.

Theorem 5.15 (Euclid)**.** *There are infinitely many primes.*

(Proof in Chapter 2.)

Divisibility tests in base 10 follow from DIC: since $10 = 9 + 1$, 10^k leaves remainder 1 when divided by 9, so n and its digit sum have the same remainder mod 9 (and mod 3). These are cleanest with congruences (Chapter 8).

5.4 Common pitfalls

- Forgetting that the remainder must satisfy $0 \leq r < b$, especially for negative a.
- Writing $a \mid b$ when you mean b/a, or using $\mid$ as an operation.
- Concluding $a \mid b$ or $a \mid c$ from $a \mid bc$; this needs a prime (Euclid's lemma, Chapter 6). For example $6 \mid 4 \cdot 9$ but $6 \nmid 4$ and $6 \nmid 9$.

5.5 Summary

- $a \mid b \iff b = ka$ for some integer k.
- TD: $a \mid b$, $b \mid c \Rightarrow a \mid c$. DIC: $a \mid b$, $a \mid c \Rightarrow a \mid bx + cy$. BBD: $a \mid b \neq 0 \Rightarrow |a| \leq |b|$.
- Division algorithm: unique q, r with $a = qb + r$, $0 \leq r < b$.
- Every $n > 1$ has a prime factor, and composite n has one at most $\sqrt{n}$; there are infinitely many primes.

Chapter 6

GCD, the Euclidean Algorithm and Unique Factorization

6.1 Greatest common divisor

Definition 6.1. Let $a, b \in \mathbb{Z}$, not both zero. The *greatest common divisor* $\gcd(a, b)$ is the largest integer d with $d \mid a$ and $d \mid b$. We set $\gcd(0, 0) = 0$.

Since 1 is always a common divisor and (by BBD) every common divisor is at most $\max(|a|, |b|)$, the gcd exists and is positive when a, b are not both 0. Simple facts: $\gcd(a, b) = \gcd(b, a) = \gcd(|a|, |b|)$, $\gcd(a, 0) = |a|$, and $\gcd(a, 1) = 1$.

Proposition 6.2 (GCD With Remainders, GCDWR)**.** *If $a = qb + r$ for integers a, b, q, r, then* $\gcd(a, b) = \gcd(b, r)$.

Proof. If d divides a and b, then by DIC $d \mid a - qb = r$. If d divides b and r, then $d \mid qb + r = a$. So a, b and b, r have exactly the same common divisors, hence the same greatest one. □

6.2 The Euclidean algorithm

Applying GCDWR repeatedly with the division algorithm shrinks the numbers until a remainder is 0; the last non-zero remainder is the gcd.

Example 6.3. Compute $\gcd(1386, 322)$.

$$\begin{aligned} 1386 &= 4 \cdot 322 + 98 \\ 322 &= 3 \cdot 98 + 28 \\ 98 &= 3 \cdot 28 + 14 \\ 28 &= 2 \cdot 14 + 0. \end{aligned}$$

So $\gcd(1386, 322) = \gcd(322, 98) = \gcd(98, 28) = \gcd(28, 14) = \gcd(14, 0) = 14$.

The remainders strictly decrease and are non-negative, so the algorithm terminates.

6.3 Bezout's lemma and the extended Euclidean algorithm

Theorem 6.4 (Bezout's Lemma / EEA)**.** *For all $a, b \in \mathbb{Z}$ there exist integers x, y with $ax + by = \gcd(a, b)$.*

The *extended Euclidean algorithm* finds such x, y. We maintain rows (x_i, y_i, r_i) with the invariant

$$ax_i + by_i = r_i.$$

Start with rows $(1, 0, a)$ and $(0, 1, b)$. Given two consecutive rows, let $q_i = \lfloor r_{i-2}/r_{i-1} \rfloor$ and set

$$\text{row}_i = \text{row}_{i-2} - q_i \cdot \text{row}_{i-1}.$$

The invariant is preserved because it is linear. Stop when $r = 0$; the previous row gives the gcd and the coefficients. (The r column is exactly the Euclidean algorithm.)

Example 6.5 (EEA for $a = 1386$, $b = 322$)**.**

x	y	r	q
1	0	1386	
0	1	322	
1	−4	98	4
−3	13	28	3
10	−43	14	3
−23	99	0	2

For instance the fourth row is $(0, 1, 322) - 3(1, -4, 98) = (-3, 13, 28)$. So $\gcd(1386, 322) = 14$ and

$$1386(10) + 322(-43) = 13860 - 13846 = 14.$$

(The last row gives $1386(-23) + 322(99) = 0$, which is useful later for generating all solutions.)

Example 6.6 (EEA for $a = 506$, $b = 391$)**.**

x	y	r	q
1	0	506	
0	1	391	
1	−1	115	1
−3	4	46	3
7	−9	23	2
−17	22	0	2

$\gcd(506, 391) = 23$ and $506(7) + 391(-9) = 3542 - 3519 = 23$.

Back-substitution

Instead of the table, one can run the Euclidean algorithm and substitute backwards: $14 = 98 - 3 \cdot 28 = 98 - 3(322 - 3 \cdot 98) = 10 \cdot 98 - 3 \cdot 322 = 10(1386 - 4 \cdot 322) - 3 \cdot 322 = 10 \cdot 1386 - 43 \cdot 322$. Same answer; the table is less error-prone.

Theorem 6.7 (GCD Characterization Theorem, GCDCT)**.** *Let $a, b \in \mathbb{Z}$ and $d \geq 0$. If $d \mid a$, $d \mid b$, and $d = ax + by$ for some integers x, y, then $d = \gcd(a, b)$.*

Proof. Let $g = \gcd(a, b)$. By DIC, $g \mid ax + by = d$, so (if $d > 0$) $g \leq d$ by BBD. But d is a common divisor, so $d \leq g$. Hence $d = g$. (If $d = 0$ then $a = b = 0$.) □

Bezout does not go backwards

$ax + by = d$ alone does not make d the gcd: $6 \cdot 1 + 4 \cdot 1 = 10$ but $\gcd(6, 4) = 2$. You also need d to be a common divisor. What *is* true is that the set of integer combinations $\{ax + by\}$ is exactly the set of multiples of $\gcd(a, b)$.

Corollary 6.8 (Common Divisor Divides GCD, CDDG)**.** *If $c \mid a$ and $c \mid b$ then $c \mid \gcd(a, b)$.*

Proof. Write $\gcd(a, b) = ax + by$ and apply DIC. □

6.4 Coprimeness

Definition 6.9. a and b are *coprime* (relatively prime) if $\gcd(a, b) = 1$.

Theorem 6.10 (Coprimeness Characterization Theorem, CCT)**.** $\gcd(a, b) = 1$ *if and only if there exist integers x, y with $ax + by = 1$.*

Proof. ($\Rightarrow$) Bezout. ($\Leftarrow$) 1 divides a and b, so by GCDCT $1 = \gcd(a, b)$. □

Proposition 6.11 (Division by GCD, DB GCD)**.** *If $d = \gcd(a, b) \neq 0$, then $\gcd(a/d,\ b/d) = 1$.*

Proof. From $ax + by = d$, divide by d: $(a/d)x + (b/d)y = 1$. Apply CCT. □

Proposition 6.12 (Coprimeness and Divisibility, CAD)**.** *If $c \mid ab$ and $\gcd(a, c) = 1$, then $c \mid b$.*

Proof. Write $ax + cy = 1$. Multiply by b: $abx + cby = b$. Now $c \mid ab$ and $c \mid c$, so by DIC $c \mid abx + cby = b$. □

Example 6.13. Prove that for any integer n, $\gcd(5n+3,\ 3n+2) = 1$: $3(5n+3) - 5(3n+2) = -1$, so $(5n + 3)(-3) + (3n + 2)(5) = 1$, and CCT applies.

6.5 Primes and Euclid's lemma

Theorem 6.14 (Euclid's Lemma / Primes and Divisibility, PAD)**.** *If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.*

Proof. Suppose $p \nmid a$. The only positive divisors of p are 1 and p, and $p \nmid a$, so $\gcd(p, a) = 1$. By CAD, $p \mid b$. □

By induction, if $p \mid a_1 a_2 \cdots a_n$ then $p \mid a_i$ for some i.

Example 6.15. $\sqrt{p}$ is irrational for every prime p: if $pq^2 = m^2$ with $\gcd(m, q) = 1$, then $p \mid m^2$ so $p \mid m$ (PAD); write $m = pk$, so $q^2 = pk^2$ and $p \mid q$, contradicting $\gcd(m, q) = 1$.

6.6 Unique factorization

Theorem 6.16 (Unique Factorization Theorem, UFT)**.** *Every integer $n > 1$ can be written as a product of primes, and this factorization is unique up to the order of the factors.*

Proof. Existence was proved by strong induction in Chapter 4. For uniqueness, suppose $n = p_1 p_2 \cdots p_k = q_1 q_2 \cdots q_\ell$ with all p_i, q_j prime; induct on k. Since $p_1 \mid q_1 \cdots q_\ell$, PAD gives $p_1 \mid q_j$ for some j, and since q_j is prime, $p_1 = q_j$. Cancel it from both sides and apply the inductive hypothesis to the shorter product. (If $k = 1$, $n = p_1$ is prime, which forces $\ell = 1$.) □

We often write the *prime factorization* as $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ with distinct primes and $\alpha_i \geq 1$; allowing $\alpha_i = 0$ lets us write two numbers over the same list of primes.

Proposition 6.17 (Divisors from Prime Factorization)**.** *If $n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$, then the positive divisors of n are exactly the numbers $p_1^{\beta_1} \cdots p_k^{\beta_k}$ with $0 \leq \beta_i \leq \alpha_i$. So n has $(\alpha_1 + 1) \cdots (\alpha_k + 1)$ positive divisors.*

Theorem 6.18 (GCD from Prime Factorization, GCDPF)**.** *If $a = \prod p_i^{\alpha_i}$ and $b = \prod p_i^{\beta_i}$ (with $\alpha_i, \beta_i \geq 0$), then*

$$\gcd(a, b) = \prod p_i^{\min(\alpha_i, \beta_i)}, \qquad \operatorname{lcm}(a, b) = \prod p_i^{\max(\alpha_i, \beta_i)}.$$

In particular $\gcd(a, b) \operatorname{lcm}(a, b) = ab$ for positive a, b.

Example 6.19. $1386 = 2 \cdot 3^2 \cdot 7 \cdot 11$ and $322 = 2 \cdot 7 \cdot 23$. Then $\gcd = 2 \cdot 7 = 14$ (agreeing with the Euclidean algorithm) and $\operatorname{lcm} = 2 \cdot 3^2 \cdot 7 \cdot 11 \cdot 23 = 31878 = 1386 \cdot 322/14$. 1386 has $(1+1)(2+1)(1+1)(1+1) = 24$ positive divisors.

> Factoring large numbers is believed to be hard, while the Euclidean algorithm is very fast (the number of steps is at most about five times the number of decimal digits). That contrast is the basis of RSA (Chapter 10).

6.7 Common pitfalls

- Concluding $d = \gcd(a, b)$ from $ax + by = d$ alone (GCDCT needs d to be a common divisor too).
- Arithmetic slips in the EEA table; always verify $ax + by = d$ at the end.
- Using PAD for a non-prime: $4 \mid 2 \cdot 6$ but $4 \nmid 2$, $4 \nmid 6$.

6.8 Summary

- GCDWR: $\gcd(a, b) = \gcd(b, r)$; iterating gives the Euclidean algorithm.
- EEA: rows (x, y, r) with $ax + by = r$; produces Bezout coefficients.
- GCDCT: a non-negative common divisor that is an integer combination is the gcd. CCT: $\gcd(a, b) = 1 \iff ax + by = 1$ is solvable.
- CAD: $c \mid ab$, $\gcd(a, c) = 1 \Rightarrow c \mid b$. PAD (Euclid's lemma): $p \mid ab \Rightarrow p \mid a$ or $p \mid b$.
- UFT: unique prime factorization; gcd/lcm via min/max of exponents.

Chapter 7

Linear Diophantine Equations

A *Diophantine equation* is an equation where we look only for integer solutions. The simplest kind is the linear equation in two variables

$$ax + by = c, \qquad a, b, c \in \mathbb{Z},$$

which is answered completely by the gcd machinery of Chapter 6.

7.1 When is there a solution?

Theorem 7.1 (Linear Diophantine Equation Theorem, Part 1 (LDET 1))**.** *Let $a, b, c \in \mathbb{Z}$ with a, b not both zero, and $d = \gcd(a, b)$. The equation $ax + by = c$ has an integer solution if and only if $d \mid c$.*

Proof. ($\Rightarrow$) If $ax_0 + by_0 = c$, then since $d \mid a$ and $d \mid b$, DIC gives $d \mid c$.

($\Leftarrow$) Write $c = kd$. By Bezout there are x', y' with $ax' + by' = d$. Then $a(kx') + b(ky') = kd = c$. □

The proof is constructive: *run the EEA, then scale by c/d.*

Example 7.2. $6x + 15y = 4$ has no integer solution because $\gcd(6, 15) = 3 \nmid 4$. (Indeed the left side is always a multiple of 3.) But $6x + 15y = 9$ does: $6(-2) + 15(1) = 3$, so $6(-6) + 15(3) = 9$.

7.2 Finding all solutions

Theorem 7.3 (Linear Diophantine Equation Theorem, Part 2 (LDET 2))**.** *Let a, b be non-zero, $d = \gcd(a, b)$, and suppose (x_0, y_0) is one integer solution of $ax + by = c$. Then the set of all integer solutions is*

$$\left\{ \left(x_0 + \frac{b}{d}n, \quad y_0 - \frac{a}{d}n \right) : n \in \mathbb{Z} \right\}.$$

Proof. These are solutions: $a\left(x_0 + \frac{b}{d}n\right) + b\left(y_0 - \frac{a}{d}n\right) = ax_0 + by_0 + \frac{ab}{d}n - \frac{ab}{d}n = c$.

Every solution has this form: let (x, y) be a solution. Subtracting, $a(x - x_0) + b(y - y_0) = 0$, so

$$\frac{a}{d}(x - x_0) = -\frac{b}{d}(y - y_0).$$

Now $\frac{b}{d}$ divides the left side, and $\gcd(\frac{a}{d}, \frac{b}{d}) = 1$ (DB GCD), so by CAD $\frac{b}{d} \mid x - x_0$. Write $x - x_0 = \frac{b}{d}n$. Substituting, $\frac{a}{d} \cdot \frac{b}{d}n = -\frac{b}{d}(y - y_0)$, and cancelling $\frac{b}{d} \neq 0$ gives $y - y_0 = -\frac{a}{d}n$. □

Divide by the gcd

The step sizes are b/d and a/d, not b and a. Using b and a produces only some of the solutions. Also note the opposite signs: when x goes up, y goes down (if $a, b > 0$).

7.3 Worked examples

Example 7.4. Find all integer solutions to $1386x + 322y = 70$.

From the EEA table in Chapter 6, $\gcd(1386, 322) = 14$ and $1386(10) + 322(-43) = 14$. Since $14 \mid 70$ (with $70 = 5 \cdot 14$), solutions exist. Multiply by 5:

$$1386(50) + 322(-215) = 70.$$

Check: $69300 - 69230 = 70$. Here $b/d = 322/14 = 23$ and $a/d = 1386/14 = 99$, so by LDET 2 all solutions are

$$x = 50 + 23n, \qquad y = -215 - 99n, \qquad n \in \mathbb{Z}.$$

Taking $n = -2$ gives the small solution $(4, -17)$: $5544 - 5474 = 70$.

Example 7.5 (A word problem with constraints)**.** Stamps cost 7 and 11 cents. In how many ways can one pay exactly 100 cents?

We need non-negative integers x, y with $7x + 11y = 100$. Euclid: $11 = 1 \cdot 7 + 4$, $7 = 1 \cdot 4 + 3$, $4 = 1 \cdot 3 + 1$. EEA table:

x	y	r	q
1	0	7	
0	1	11	
1	0	7	0
-1	1	4	1
2	-1	3	1
-3	2	1	1

(The first step with $q = 0$ just swaps the order; you may also simply start with the larger number first.) So $7(-3) + 11(2) = 1$ and $7(-300) + 11(200) = 100$. All integer solutions:

$$x = -300 + 11n, \qquad y = 200 - 7n.$$

Now impose $x \geq 0$: $n \geq 300/11 \approx 27.3$, so $n \geq 28$; and $y \geq 0$: $n \leq 200/7 \approx 28.6$, so $n \leq 28$. Hence $n = 28$, giving the unique answer $(x, y) = (8, 4)$: $56 + 44 = 100$.

Example 7.6. Find all solutions of $15x - 21y = 39$ with x between -10 and 10.

$\gcd(15, 21) = 3$ and $3 \mid 39$. A Bezout relation: $15 \cdot 3 - 21 \cdot 2 = 45 - 42 = 3$. Times 13: $15(39) - 21(26) = 39$, so $(x_0, y_0) = (39, 26)$ solves $15x - 21y = 39$. Writing the equation as $15x + (-21)y = 39$, LDET 2 gives $x = 39 + \frac{-21}{3}n = 39 - 7n$ and $y = 26 - \frac{15}{3}n = 26 - 5n$. For $-10 \leq 39 - 7n \leq 10$ we need $29/7 \leq n \leq 7$, i.e. $n \in \{5, 6, 7\}$, giving $(4, 1)$, $(-3, -4)$, $(-10, -9)$. Check $(4, 1)$: $60 - 21 = 39$.

Geometrically, the integer solutions of $ax + by = c$ are the lattice points on a line; consecutive ones are separated by the vector $(b/d, -a/d)$.

7.4 Connection with congruences

$ax + by = c$ has a solution in x exactly when $ax \equiv c$ has a solution modulo b (Chapter 8): the y just absorbs a multiple of b. Thus LDET 1 and 2 will give us the theory of linear congruences.

7.5 Common pitfalls

- Forgetting to check $d \mid c$ before trying to solve.
- Using $x_0 + bn$ instead of $x_0 + \frac{b}{d}n$.
- Sign errors when a or b is negative; rewrite the equation as $ax + by = c$ with the signs inside a, b and apply the formula literally.
- Not verifying the particular solution by substitution.

7.6 Summary

- LDET 1: $ax + by = c$ is solvable $\iff \gcd(a, b) \mid c$.
- To find a particular solution: EEA for $ax + by = d$, then multiply by c/d.
- LDET 2: all solutions are $x = x_0 + \frac{b}{d}n$, $y = y_0 - \frac{a}{d}n$, $n \in \mathbb{Z}$.
- Constraints (non-negativity, ranges) become inequalities on the parameter n.

Chapter 8

Congruences and Modular Arithmetic

8.1 Congruence

Definition 8.1. Let $m \in \mathbb{N}$. Integers a and b are *congruent modulo* m, written $a \equiv b \pmod{m}$, if $m \mid (a-b)$.

Example 8.2. $17 \equiv 2 \pmod{5}$ since $5 \mid 15$; $-3 \equiv 9 \pmod{4}$ since $4 \mid -12$; $10 \not\equiv 3 \pmod{4}$ since $4 \nmid 7$. $a \equiv 0 \pmod{m}$ means $m \mid a$.

Proposition 8.3 (Congruence is an Equivalence Relation, CER)**.** *For all $a, b, c \in \mathbb{Z}$:*

1. *(reflexive) $a \equiv a \pmod{m}$;*
2. *(symmetric) $a \equiv b \Rightarrow b \equiv a \pmod{m}$;*
3. *(transitive) $a \equiv b$ and $b \equiv c \Rightarrow a \equiv c \pmod{m}$.*

Proof. (1) $m \mid 0$. (2) If $m \mid a-b$ then $m \mid -(a-b) = b-a$. (3) $m \mid (a-b)+(b-c) = a-c$ by DIC. □

Theorem 8.4 (Congruence Add and Multiply, CAM)**.** *If $a \equiv a'$ and $b \equiv b' \pmod{m}$, then*

$$a+b \equiv a'+b', \qquad a-b \equiv a'-b', \qquad ab \equiv a'b' \pmod{m}.$$

Consequently $a^k \equiv (a')^k \pmod{m}$ for every $k \in \mathbb{N}$ (Congruence Power, CP).

Proof. For products: $ab - a'b' = a(b-b') + b'(a-a')$, and m divides both terms, so DIC applies. The power rule follows by induction on k. □

Division is not automatic

We can add, subtract, multiply and take powers, but *not* divide freely: $2 \cdot 3 \equiv 2 \cdot 8 \pmod{10}$ but $3 \not\equiv 8 \pmod{10}$.

Proposition 8.5 (Congruence Divide, CD)**.** *If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then $a \equiv b \pmod{m}$.*

Proof. $m \mid c(a-b)$ and $\gcd(m, c) = 1$, so by CAD $m \mid a-b$. □

Proposition 8.6 (Congruent iff Same Remainder, CISR)**.** *$a \equiv b \pmod{m}$ if and only if a and b leave the same remainder on division by m.*

Proof. Write $a = q_1 m + r_1$, $b = q_2 m + r_2$ with $0 \le r_i < m$. Then $a - b = (q_1 - q_2)m + (r_1 - r_2)$, so $m \mid a-b \iff m \mid r_1 - r_2$, and since $|r_1 - r_2| < m$, this happens iff $r_1 = r_2$. □

8.2 Computing with congruences

Example 8.7. Find the remainder when 2^{100} is divided by 7. Since $2^3 = 8 \equiv 1 \pmod 7$,

$$2^{100} = 2^{99} \cdot 2 = (2^3)^{33} \cdot 2 \equiv 1^{33} \cdot 2 = 2 \pmod 7.$$

The remainder is 2.

Example 8.8. Find the last digit of 7^{2022} (i.e. the remainder mod 10). $7^2 = 49 \equiv -1 \pmod{10}$, so $7^4 \equiv 1$. Then $7^{2022} = (7^4)^{505} \cdot 7^2 \equiv 49 \equiv 9 \pmod{10}$. Last digit: 9.

Example 8.9 (Repeated squaring)**.** Compute 3^{45} mod 23. Squaring: $3^1 = 3$, $3^2 = 9$, $3^4 = 81 \equiv 12$, $3^8 \equiv 144 \equiv 6$, $3^{16} \equiv 36 \equiv 13$, $3^{32} \equiv 169 \equiv 8 \pmod{23}$. With $45 = 32 + 8 + 4 + 1$:

$$3^{45} \equiv 8 \cdot 6 \cdot 12 \cdot 3 = 1728 \equiv 1728 - 75 \cdot 23 = 1728 - 1725 = 3 \pmod{23}.$$

(Consistent with $3^{11} \equiv 1 \pmod{23}$, since then $3^{45} = (3^{11})^4 \cdot 3 \equiv 3$.)

Example 8.10 (Divisibility by 9 and 11)**.** Write $n = \sum_{i=0}^{k} d_i 10^i$. Since $10 \equiv 1 \pmod 9$, $n \equiv \sum d_i \pmod 9$. Since $10 \equiv -1 \pmod{11}$, $n \equiv \sum (-1)^i d_i \pmod{11}$. For example 918082: digit sum $28 \equiv 1 \pmod 9$, so $9 \nmid 918082$; alternating sum from the right $2-8+0-8+1-9 = -22 \equiv 0 \pmod{11}$, so $11 \mid 918082$.

Example 8.11. Show $x^2 \equiv 3 \pmod 4$ has no solutions (hence $x^2 + y^2 = 4k + 3$ is impossible). Every x is congruent to $0, 1, 2, 3$ mod 4, and then $x^2 \equiv 0, 1, 0, 1$. Never 3.

8.3 Congruence classes and $\mathbb{Z}_m$

Definition 8.12. The *congruence class* of a modulo m is $[a] = \{x \in \mathbb{Z} : x \equiv a \pmod m\} = \{a + km : k \in \mathbb{Z}\}$. The set of classes is $\mathbb{Z}_m = \{[0], [1], \ldots, [m-1]\}$, with operations $[a] + [b] = [a+b]$ and $[a] \cdot [b] = [ab]$.

By CAM these operations are well defined (independent of the representatives chosen), and by CISR the m listed classes are distinct and cover $\mathbb{Z}$. $\mathbb{Z}_m$ satisfies the usual rules of arithmetic (commutative, associative, distributive, identities [0] and [1], additive inverses $-[a] = [-a]$).

Example 8.13 (Multiplication table of $\mathbb{Z}_6$, non-zero part)**.**

·	1	2	3	4	5
1	1	2	3	4	5
2	2	4	0	2	4
3	3	0	3	0	3
4	4	2	0	4	2
5	5	4	3	2	1

Note $[2][3] = [0]$: $\mathbb{Z}_6$ has zero divisors, and only $[1], [5]$ have multiplicative inverses.

8.4 Inverses in $\mathbb{Z}_m$

Definition 8.14. $[b]$ is a *multiplicative inverse* of $[a]$ in $\mathbb{Z}_m$ if $[a][b] = [1]$, i.e. $ab \equiv 1 \pmod m$.

Theorem 8.15 (Inverses in $\mathbb{Z}_m$, INV $\mathbb{Z}_m$)**.** *$[a]$ has a multiplicative inverse in $\mathbb{Z}_m$ if and only if $\gcd(a, m) = 1$, and then the inverse is unique. In particular, if p is prime, every non-zero element of $\mathbb{Z}_p$ has an inverse ($\mathbb{Z}_p$ is a field).*

Proof. $ab \equiv 1 \pmod m \iff ab + mk = 1$ for some $k \iff \gcd(a, m) = 1$ (CCT). Uniqueness: if $ab \equiv ab' \equiv 1$, then multiplying $ab \equiv ab'$ by b gives $b \equiv b'$. □

Example 8.16. Find $[17]^{-1}$ in $\mathbb{Z}_{37}$. EEA on 37 and 17: $37 = 2 \cdot 17 + 3$, $17 = 5 \cdot 3 + 2$, $3 = 1 \cdot 2 + 1$.

x	y	r	q
1	0	37	
0	1	17	
1	−2	3	2
−5	11	2	5
6	−13	1	1

So $37(6) + 17(-13) = 1$, giving $17 \cdot (-13) \equiv 1 \pmod{37}$. Thus $[17]^{-1} = [-13] = [24]$. Check: $17 \cdot 24 = 408 = 11 \cdot 37 + 1$.

8.5 Linear congruences

Theorem 8.17 (Linear Congruence Theorem, LCT)**.** *Let $d = \gcd(a, m)$. The congruence $ax \equiv c \pmod m$ has a solution if and only if $d \mid c$. In that case, if x_0 is one solution, the complete solution is*

$$x \equiv x_0 \pmod{\tfrac{m}{d}},$$

equivalently $x \equiv x_0,\ x_0 + \frac{m}{d},\ \ldots,\ x_0 + (d-1)\frac{m}{d} \pmod m$, which are d distinct classes mod m.

Proof. $ax \equiv c \pmod m \iff ax + my = c$ for some $y \in \mathbb{Z}$. Apply LDET 1 and LDET 2: the x-values of solutions are exactly $x_0 + \frac{m}{d}n$, $n \in \mathbb{Z}$. □

Example 8.18. Solve $17x \equiv 5 \pmod{37}$. Since gcd = 1, multiply by the inverse 24: $x \equiv 24 \cdot 5 = 120 \equiv 120 - 111 = 9 \pmod{37}$. Check: $17 \cdot 9 = 153 = 4 \cdot 37 + 5$.

Example 8.19. Solve $14x \equiv 10 \pmod{40}$. Here $d = \gcd(14, 40) = 2$ and $2 \mid 10$, so there are solutions. Divide everything by 2: $7x \equiv 5 \pmod{20}$. Since $7 \cdot 3 = 21 \equiv 1$, $7^{-1} \equiv 3$ and $x \equiv 15 \pmod{20}$. As classes mod 40: $x \equiv 15$ or $35 \pmod{40}$. Check: $14 \cdot 15 = 210 = 5 \cdot 40 + 10$ and $14 \cdot 35 = 490 = 12 \cdot 40 + 10$.

Example 8.20. $6x \equiv 4 \pmod 9$ has no solution, since $\gcd(6, 9) = 3 \nmid 4$.

Dividing a congruence

When you divide $ax \equiv c \pmod m$ by $d = \gcd(a, m)$, you must divide the modulus as well: $ax \equiv c \pmod m \iff \frac{a}{d}x \equiv \frac{c}{d} \pmod{\frac{m}{d}}$.

8.6 Common pitfalls

- Cancelling a factor that is not coprime to m.
- Forgetting that negative representatives are allowed and often convenient ($7^2 \equiv -1 \pmod{10}$).
- Reducing exponents modulo m: $a^k \not\equiv a^{k \bmod m}$ in general. (Exponents reduce modulo $p - 1$ via Fermat, Chapter 9.)
- Giving the solution of a linear congruence only mod m/d when asked for classes mod m, or vice versa.

8.7 Summary

- $a \equiv b \pmod{m} \iff m \mid a - b \iff$ same remainder mod m.
- CAM/CP: congruences may be added, subtracted, multiplied, raised to powers; division only by numbers coprime to m (CD).
- $\mathbb{Z}_m$ is the set of classes; $[a]$ is invertible iff $\gcd(a, m) = 1$; find inverses with the EEA.
- LCT: $ax \equiv c \pmod{m}$ is solvable iff $d = \gcd(a, m) \mid c$, with a unique solution mod m/d (d solutions mod m).

Chapter 9

Fermat's Little Theorem and the Chinese Remainder Theorem

9.1 Fermat's Little Theorem

Theorem 9.1 (Fermat's Little Theorem, FLT)**.** *If p is prime and $p \nmid a$, then*

$$a^{p-1} \equiv 1 \pmod{p}.$$

Proof. Consider the $p-1$ numbers $a, 2a, 3a, \ldots, (p-1)a$. None is $\equiv 0 \pmod{p}$ (PAD, since $p \nmid a$ and $p \nmid i$). They are pairwise incongruent: if $ia \equiv ja$ then $i \equiv j$ by CD since $\gcd(a,p) = 1$, and $1 \le i, j \le p-1$ forces $i = j$. So, modulo p, they are a rearrangement of $1, 2, \ldots, p-1$. Multiplying all of them,

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}.$$

Since $\gcd((p-1)!, p) = 1$, we may cancel (CD) to get $a^{p-1} \equiv 1 \pmod{p}$. □

Corollary 9.2. *For every prime p and every integer a, $a^p \equiv a \pmod{p}$. (If $p \mid a$ both sides are 0; otherwise multiply FLT by a.)*

Reducing exponents

If $p \nmid a$ and $k \equiv r \pmod{p-1}$, then $a^k \equiv a^r \pmod{p}$. So exponents may be reduced *modulo* $p-1$, not modulo p. Also a^{p-2} is an inverse of a modulo p.

Example 9.3. Remainder of 2^{1000} divided by 13: by FLT $2^{12} \equiv 1$. Since $1000 = 12 \cdot 83 + 4$, $2^{1000} = (2^{12})^{83} \cdot 2^4 \equiv 16 \equiv 3 \pmod{13}$.

Example 9.4. Remainder of 3^{2025} divided by 31: $3^{30} \equiv 1$ and $2025 = 30 \cdot 67 + 15$, so $3^{2025} \equiv 3^{15}$. Now $3^3 = 27 \equiv -4$, so $3^{15} \equiv (-4)^5 = -1024$. Since $1024 = 33 \cdot 31 + 1$, $-1024 \equiv -1 \equiv 30$. The remainder is 30.

Example 9.5. Solve $x^{39} \equiv 3 \pmod{13}$. Clearly $13 \nmid x$. By FLT $x^{12} \equiv 1$, so $x^{39} = (x^{12})^3 x^3 \equiv x^3$. Now test cubes mod 13: $1, 8, 1, 12, 8, 8, 5, 5, 1, 12, 5, 12$ for $x = 1, \ldots, 12$. None is 3, so there is no solution.

Example 9.6 (FLT as a compositeness test)**.** If $a^{n-1} \not\equiv 1 \pmod{n}$ for some a with $\gcd(a,n) = 1$, then n is composite. E.g. $2^{14} = 16384 = 1092 \cdot 15 + 4$, so $2^{14} \equiv 4 \not\equiv 1 \pmod{15}$ and 15 is not prime. (The converse of FLT fails: $341 = 11 \cdot 31$ satisfies $2^{340} \equiv 1 \pmod{341}$.)

FLT generalizes to Euler's theorem: if $\gcd(a, n) = 1$ then $a^{\varphi(n)} \equiv 1 \pmod{n}$, where $\varphi(n)$ counts $1 \le k \le n$ with $\gcd(k, n) = 1$. For $n = pq$ with distinct primes, $\varphi(n) = (p-1)(q-1)$, the number used in RSA.

9.2 The Chinese Remainder Theorem

Theorem 9.7 (Chinese Remainder Theorem, CRT)**.** *Let $m_1, m_2 \in \mathbb{N}$ with $\gcd(m_1, m_2) = 1$. For any integers a_1, a_2, the system*

$$x \equiv a_1 \pmod{m_1}, \qquad x \equiv a_2 \pmod{m_2}$$

has a solution, and if x_0 is one solution, the complete solution is $x \equiv x_0 \pmod{m_1 m_2}$.

Proof. Existence: solutions of the first congruence are $x = a_1 + m_1 k$. Substituting into the second, we need $m_1 k \equiv a_2 - a_1 \pmod{m_2}$, which is solvable by LCT because $\gcd(m_1, m_2) = 1$.

Uniqueness: if x_0 and x are solutions then $m_1 \mid x - x_0$ and $m_2 \mid x - x_0$. Write $x - x_0 = m_1 t$; then $m_2 \mid m_1 t$ and by CAD $m_2 \mid t$, so $m_1 m_2 \mid x - x_0$. Conversely anything $\equiv x_0 \pmod{m_1 m_2}$ is clearly a solution. □

By induction, the theorem extends to any number of pairwise coprime moduli $m_1, \ldots, m_k$: the system has a unique solution modulo $m_1 m_2 \cdots m_k$ (Generalized CRT).

9.2.1 Method 1: substitution

Example 9.8. Solve $x \equiv 2 \pmod 3$, $x \equiv 3 \pmod 5$, $x \equiv 2 \pmod 7$.

From the first, $x = 2+3k$. Then $2+3k \equiv 3 \pmod 5$, so $3k \equiv 1$; since $3 \cdot 2 = 6 \equiv 1$, $k \equiv 2 \pmod 5$, i.e. $k = 2 + 5j$ and $x = 8 + 15j$. Then $8 + 15j \equiv 2 \pmod 7$; as $8 \equiv 1$ and $15 \equiv 1$, this is $1 + j \equiv 2$, so $j \equiv 1 \pmod 7$, $j = 1 + 7t$, and $x = 23 + 105t$.

Answer: $x \equiv 23 \pmod{105}$. Check: $23 = 7 \cdot 3 + 2 = 4 \cdot 5 + 3 = 3 \cdot 7 + 2$.

9.2.2 Method 2: the constructive formula

Let $M = m_1 \cdots m_k$ and $M_i = M/m_i$. Since $\gcd(M_i, m_i) = 1$, let y_i be an inverse of M_i modulo m_i. Then

$$x \equiv a_1 M_1 y_1 + a_2 M_2 y_2 + \cdots + a_k M_k y_k \pmod{M}$$

works, because modulo m_i every term except the i-th vanishes, and the i-th is $a_i M_i y_i \equiv a_i$.

Example 9.9. Same system: $M = 105$; $M_1 = 35 \equiv 2 \pmod 3$, inverse $y_1 = 2$; $M_2 = 21 \equiv 1 \pmod 5$, $y_2 = 1$; $M_3 = 15 \equiv 1 \pmod 7$, $y_3 = 1$. Then

$$x \equiv 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 = 140 + 63 + 30 = 233 \equiv 23 \pmod{105}.$$

Example 9.10. Solve $x \equiv 3 \pmod 4$, $x \equiv 5 \pmod 9$, $x \equiv 11 \pmod{25}$ (pairwise coprime).

Start with the largest modulus: $x = 11 + 25k$. Mod 9: $11 \equiv 2$, $25 \equiv 7$, so $2 + 7k \equiv 5$, i.e. $7k \equiv 3 \pmod 9$. Since $7 \cdot 4 = 28 \equiv 1$, $k \equiv 12 \equiv 3 \pmod 9$. So $k = 3 + 9j$ and $x = 86 + 225j$. Mod 4: $86 \equiv 2$, $225 \equiv 1$, so $2 + j \equiv 3$, $j \equiv 1 \pmod 4$, giving $x = 311 + 900t$.

Answer: $x \equiv 311 \pmod{900}$. Check: $311 = 77 \cdot 4 + 3 = 34 \cdot 9 + 5 = 12 \cdot 25 + 11$.

9.2.3 Non-coprime moduli

If the moduli are not coprime, the system may have no solution or a solution modulo $\text{lcm}(m_1, m_2)$. Substitution still works.

Example 9.11. $x \equiv 3 \pmod 6$, $x \equiv 5 \pmod 8$: $x = 3+6k$, need $6k \equiv 2 \pmod 8$. $\gcd(6,8) = 2 \mid 2$, so divide: $3k \equiv 1 \pmod 4$, $k \equiv 3 \pmod 4$. Then $x = 21+24t$, i.e. $x \equiv 21 \pmod{24}$ ($24 = \text{lcm}(6,8)$).

$x \equiv 1 \pmod 6$, $x \equiv 2 \pmod 4$ has no solution: the first forces x odd and the second forces x even.

9.3 Splitting the modulus

Theorem 9.12 (Splitting Modulus Theorem, SMT)**.** *If* $\gcd(m_1, m_2) = 1$*, then*

$$a \equiv b \pmod{m_1 m_2} \iff a \equiv b \pmod{m_1} \text{ and } a \equiv b \pmod{m_2}.$$

Proof. ($\Rightarrow$) by TD. ($\Leftarrow$) $m_1 \mid a-b$ and $m_2 \mid a-b$ with coprime moduli, so $m_1 m_2 \mid a-b$ (as in the uniqueness part of CRT). □

This lets us solve a hard congruence modulo a composite number by solving it modulo each prime-power factor and recombining with CRT.

Example 9.13. Solve $x^2 \equiv 4 \pmod{35}$. By SMT this is equivalent to $x^2 \equiv 4 \pmod 5$ and $x^2 \equiv 4 \pmod 7$. Modulo a prime p, $x^2 \equiv 4 \iff p \mid (x-2)(x+2) \iff x \equiv \pm 2$ (PAD). So $x \equiv 2$ or $3 \pmod 5$ and $x \equiv 2$ or $5 \pmod 7$. Four CRT systems:

mod 5	mod 7	mod 35
2	2	2
2	5	12
3	2	23
3	5	33

So $x \equiv 2, 12, 23, 33 \pmod{35}$. Check: $12^2 = 144 = 4 \cdot 35 + 4$ and $23^2 = 529 = 15 \cdot 35 + 4$. A quadratic can have more than two roots modulo a composite!

Example 9.14. Find $5^{100} \bmod 77$. By FLT, $5^{10} \equiv 1 \pmod{11}$, so $5^{100} \equiv 1 \pmod{11}$; and $5^6 \equiv 1 \pmod 7$, $100 = 6 \cdot 16 + 4$, so $5^{100} \equiv 5^4 = 625 \equiv 625 - 623 = 2 \pmod 7$. Solve $x \equiv 1 \pmod{11}$, $x \equiv 2 \pmod 7$: $x = 1+11k$, $1+4k \equiv 2$, $4k \equiv 1$, $k \equiv 2 \pmod 7$. So $x = 23$, and $5^{100} \equiv 23 \pmod{77}$.

9.4 Common pitfalls

- Reducing exponents mod p instead of mod $p-1$, or applying FLT when $p \mid a$.
- Applying CRT's "unique mod $m_1 m_2$" conclusion to non-coprime moduli.
- Forgetting to check the final answer in every congruence.

9.5 Summary

- FLT: p prime, $p \nmid a \Rightarrow a^{p-1} \equiv 1 \pmod{p}$; always $a^p \equiv a$.
- CRT: coprime moduli $\Rightarrow$ unique solution modulo the product. Solve by substitution or by $x = \sum a_i M_i y_i$.
- SMT: a congruence mod $m_1 m_2$ (coprime) is the pair of congruences mod m_1 and mod m_2; combine solutions with CRT.

Chapter 10

RSA Public-Key Cryptography

RSA (Rivest, Shamir, Adleman, 1977) is the payoff of the number theory so far: it uses the Euclidean algorithm, modular inverses, Fermat's Little Theorem and the splitting modulus theorem.

10.1 Public-key cryptography

In *symmetric* (private-key) cryptography, sender and receiver share a secret key used both to encrypt and decrypt; the problem is agreeing on that key securely. In *public-key* cryptography, each receiver has two keys:

- a *public key*, published to everyone and used to *encrypt* messages to her;
- a *private key*, kept secret and used to *decrypt*.

Security rests on the belief that computing the private key from the public key is infeasible. For RSA, this is (essentially) the difficulty of factoring a large $n = pq$.

10.2 The RSA scheme

RSA key generation

1. Choose two distinct large primes p and q (hundreds of digits in practice).
2. Compute $n = pq$ and $\varphi = (p-1)(q-1)$.
3. Choose e with $1 < e < \varphi$ and $\gcd(e, \varphi) = 1$.
4. Compute d with $1 < d < \varphi$ and $ed \equiv 1 \pmod{\varphi}$ (EEA).
5. Public key: (e, n). Private key: (d, n). Keep p, q, φ secret.

Encryption and decryption

A message is an integer M with $0 \leq M < n$.

- Encrypt: $C \equiv M^e \pmod{n}$, with $0 \leq C < n$.
- Decrypt: $R \equiv C^d \pmod{n}$, with $0 \leq R < n$. Then $R = M$.

Why it is (believed) secure: an eavesdropper knows n, e and C. To find d she needs φ; but knowing n and $\varphi = n - (p+q) + 1$ is equivalent to knowing $p+q$ and pq, hence p and q (roots of $t^2 - (p+q)t + pq$). So recovering d this way is as hard as factoring n.

All the steps for the legitimate users are fast: primality testing, the EEA, and modular exponentiation by repeated squaring each take a number of steps polynomial in the number of digits.

10.3 A small worked example

Example 10.1 (Key generation)**.** Take $p = 11$, $q = 13$. Then $n = 143$ and $\varphi = 10 \cdot 12 = 120$. Choose $e = 7$; $\gcd(7, 120) = 1$ since $120 = 2^3 \cdot 3 \cdot 5$. To find d, run the EEA on 120 and 7:

x	y	r	q
1	0	120	
0	1	7	
1	-17	1	17

So $120(1) + 7(-17) = 1$, i.e. $7 \cdot (-17) \equiv 1 \pmod{120}$, and $d \equiv -17 \equiv 103 \pmod{120}$. Check: $7 \cdot 103 = 721 = 6 \cdot 120 + 1$.

Public key $(e, n) = (7, 143)$; private key $(d, n) = (103, 143)$.

Example 10.2 (Encryption)**.** Encrypt $M = 9$: compute 9^7 mod 143 by repeated squaring.

$$\begin{aligned} 9^2 &= 81, \\ 9^4 &= 81^2 = 6561 = 45 \cdot 143 + 126 \equiv 126 \equiv -17, \\ 9^7 &= 9^4 \cdot 9^2 \cdot 9 \equiv (-17)(81)(9) = (-17)(729). \end{aligned}$$

Since $729 = 5 \cdot 143 + 14$, this is $(-17)(14) = -238 \equiv -238 + 2 \cdot 143 = 48$. So the ciphertext is $C = 48$.

Example 10.3 (Decryption)**.** Compute 48^{103} mod 143. The receiver knows p, q, so she may use SMT and FLT.

Mod 11*:* $48 \equiv 4$. By FLT exponents reduce mod 10: $103 \equiv 3$, so $48^{103} \equiv 4^3 = 64 \equiv 9 \pmod{11}$.

Mod 13*:* $48 \equiv 9$. Exponents reduce mod 12: $103 = 8 \cdot 12 + 7$, so we need 9^7. $9^2 = 81 \equiv 3$, $9^4 \equiv 9$, $9^7 = 9^4 \cdot 9^2 \cdot 9 \equiv 9 \cdot 3 \cdot 9 = 243 = 18 \cdot 13 + 9 \equiv 9 \pmod{13}$.

So $R \equiv 9 \pmod{11}$ and $R \equiv 9 \pmod{13}$; by CRT/SMT, $R \equiv 9 \pmod{143}$. We recover $M = 9$. (Without knowing p, q, the same result follows from repeated squaring mod 143 directly, just with more work.)

10.4 Proof of correctness

Theorem 10.4 (RSA works)**.** *With n, e, d as above and $0 \le M < n$, if $C \equiv M^e \pmod{n}$ and $R \equiv C^d \pmod{n}$ with $0 \le R < n$, then $R = M$.*

Proof. Since $ed \equiv 1 \pmod{\varphi}$, write $ed = 1 + k(p-1)(q-1)$ for some integer $k \ge 0$. Then

$$R \equiv C^d \equiv M^{ed} \pmod{n}.$$

We show $M^{ed} \equiv M \pmod{p}$.

- If $p \nmid M$: by FLT $M^{p-1} \equiv 1 \pmod{p}$, so $M^{ed} = M \cdot \left(M^{p-1}\right)^{k(q-1)} \equiv M \cdot 1 = M \pmod{p}$.
- If $p \mid M$: both M^{ed} and M are $\equiv 0 \pmod{p}$.

By the same argument, $M^{ed} \equiv M \pmod{q}$. Since $p \neq q$ are primes, $\gcd(p, q) = 1$, and SMT gives $M^{ed} \equiv M \pmod{pq}$. So $R \equiv M \pmod{n}$, and since both R and M lie in $\{0, 1, \ldots, n-1\}$, CISR gives $R = M$. □

Why $p \neq q$ and $\gcd(e, \varphi) = 1$

$\gcd(e, \varphi) = 1$ is exactly what makes the inverse d exist (INV $\mathbb{Z}_m$). Distinctness of p and q is used when we apply SMT; with $p = q$, $n = p^2$ is trivially factored anyway.

10.5 Digital signatures

RSA also gives signatures. To sign M, Alice computes $S \equiv M^d \pmod{n}$ with her *private* key. Anyone can verify by checking $S^e \equiv M \pmod{n}$ with her public key; only someone knowing d could have produced S. Correctness is the same theorem with the roles of e and d swapped (since $ed = de$).

Example 10.5. With Alice's keys above, signing $M = 2$: $S \equiv 2^{103} \pmod{143}$. Mod 11: $2^3 = 8$. Mod 13: $2^7 = 128 = 9 \cdot 13 + 11 \equiv 11$. Solve $S \equiv 8 \pmod{11}$, $S \equiv 11 \pmod{13}$: $S = 11 + 13k$, $11 + 2k \equiv 8 \pmod{11}$, so $2k \equiv 8$ and $k \equiv 4$, giving $S = 63$. Verify: $63^7 \bmod 143$. Mod 11: $63 \equiv 8$, and $8^7 \equiv 8^{-3} \equiv 2$ (as $8^3 = 512 = 46 \cdot 11 + 6$ and $6 \cdot 2 = 12 \equiv 1$). Mod 13: $63 \equiv 11 \equiv -2$, $(-2)^7 = -128 \equiv -128 + 130 = 2$. So $63^7 \equiv 2$, as required.

10.6 Common pitfalls

- Inverting e modulo n instead of modulo $\varphi = (p-1)(q-1)$.
- Choosing e with $\gcd(e, \varphi) \neq 1$ (e.g. any even e, since φ is even).
- Forgetting the case $p \mid M$ in the correctness proof.
- Arithmetic errors in repeated squaring; reduce after every multiplication and use negative representatives to keep numbers small.

10.7 Summary

- Keys: $n = pq$, $\varphi = (p-1)(q-1)$, $\gcd(e, \varphi) = 1$, $ed \equiv 1 \pmod{\varphi}$; public (e, n), private (d, n).
- $C \equiv M^e$, $M \equiv C^d \pmod{n}$.
- Correctness: FLT modulo p and q separately, then SMT.
- Security relies on the difficulty of factoring n; legitimate operations are fast.

Chapter 11

Complex Numbers

11.1 Definition and arithmetic

Definition 11.1. A *complex number* is an expression $z = x + yi$ with $x, y \in \mathbb{R}$, where $i^2 = -1$. $x = \mathrm{Re}(z)$ is the *real part* and $y = \mathrm{Im}(z)$ the *imaginary part* (a real number). The set of complex numbers is $\mathbb{C}$. Two complex numbers are equal iff their real parts are equal and their imaginary parts are equal.

Arithmetic follows the usual rules with $i^2 = -1$:

$$(a + bi) + (c + di) = (a + c) + (b + d)i,$$
$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

Powers of i cycle with period 4: $i, -1, -i, 1, i, \ldots$; e.g. $i^{103} = i^{100} i^3 = -i$.

Example 11.2. $(3 + 2i)(1 - 4i) = 3 - 12i + 2i - 8i^2 = 3 + 8 - 10i = 11 - 10i$.

Proposition 11.3. *Every non-zero $z = x + yi$ has a multiplicative inverse $z^{-1} = \dfrac{x - yi}{x^2 + y^2}$. With the operations above, $\mathbb{C}$ is a field.*

11.2 Conjugate and modulus

Definition 11.4. The *conjugate* of $z = x + yi$ is $\overline{z} = x - yi$. The *modulus* is $|z| = \sqrt{x^2 + y^2}$, a non-negative real number.

Proposition 11.5 (Properties of Conjugate, PCJ)**.** *For $z, w \in \mathbb{C}$:*

1. *$\overline{\overline{z}} = z$, $\overline{z + w} = \overline{z} + \overline{w}$, $\overline{zw} = \overline{z}\,\overline{w}$, $\overline{z/w} = \overline{z}/\overline{w}$ $(w \neq 0)$;*
2. *$z + \overline{z} = 2\,\mathrm{Re}(z)$ and $z - \overline{z} = 2i\,\mathrm{Im}(z)$;*
3. *$z \in \mathbb{R} \iff z = \overline{z}$, and z is purely imaginary iff $\overline{z} = -z$;*
4. *$z\overline{z} = |z|^2$.*

Proposition 11.6 (Properties of Modulus, PM)**.**

1. *$|z| = 0 \iff z = 0$; $|\overline{z}| = |z|$;*
2. *$|zw| = |z|\,|w|$ and $|z/w| = |z| \,/\, |w|$;*

3. (Triangle inequality) $|z+w| \le |z| + |w|$.

Proof of the triangle inequality. $|z+w|^2 = (z+w)(\overline{z}+\overline{w}) = |z|^2 + z\overline{w} + \overline{z\overline{w}} + |w|^2 = |z|^2 + 2\,\mathrm{Re}(z\overline{w}) + |w|^2 \le |z|^2 + 2\,|z|\,|w| + |w|^2 = (|z| + |w|)^2$, using $\mathrm{Re}(u) \le |u|$. Take square roots. □

Division is done by multiplying numerator and denominator by the conjugate of the denominator.

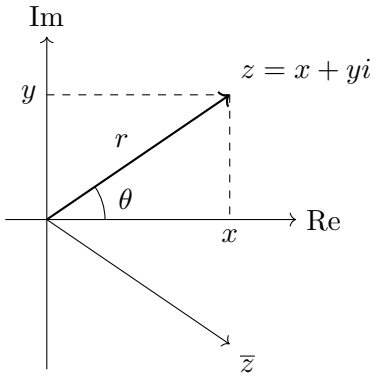
Example 11.7. $\dfrac{2+3i}{1-i} = \dfrac{(2+3i)(1+i)}{(1-i)(1+i)} = \dfrac{2+2i+3i+3i^2}{2} = \dfrac{-1+5i}{2} = -\dfrac{1}{2} + \dfrac{5}{2}i.$

Example 11.8 (Square roots algebraically)**.** Solve $z^2 = -3+4i$. Write $z = a+bi$: $a^2 - b^2 = -3$ and $2ab = 4$. Also $|z|^2 = |-3+4i| = 5$, so $a^2 + b^2 = 5$. Adding, $a^2 = 1$; subtracting, $b^2 = 4$. Since $ab = 2 > 0$, a, b have the same sign: $z = \pm(1+2i)$. Check: $(1+2i)^2 = 1 + 4i - 4 = -3 + 4i$.

Example 11.9 (Complex quadratic)**.** Solve $z^2 + 2z + (1-2i) = 0$. The quadratic formula holds over $\mathbb{C}$: discriminant $\Delta = 4 - 4(1-2i) = 8i$. A square root of $8i$ is $2+2i$ since $(2+2i)^2 = 4 + 8i - 4 = 8i$. So $z = \frac{-2 \pm (2+2i)}{2}$, i.e. $z = i$ or $z = -2 - i$. Check: $i^2 + 2i + 1 - 2i = 0$.

11.3 The complex plane and polar form

Identify $z = x + yi$ with the point (x, y) in the *complex plane* (Argand diagram): the horizontal axis is the real axis and the vertical axis the imaginary axis. $|z|$ is the distance from 0, $\overline{z}$ is the reflection in the real axis, and addition is vector addition.



Definition 11.10. If $z \neq 0$ has modulus $r = |z|$ and makes angle θ with the positive real axis, then

$$z = r(\cos\theta + i\sin\theta),$$

the *polar form* of z. Any such θ is an *argument* of z; arguments are determined up to multiples of 2π. We abbreviate $\cos\theta + i\sin\theta = \mathrm{cis}\,\theta$ and also write it $e^{i\theta}$.

To convert: $r = \sqrt{x^2 + y^2}$, and θ satisfies $\cos\theta = x/r$, $\sin\theta = y/r$ (choose the quadrant from the signs of x and y; $\tan\theta = y/x$ alone is ambiguous).

Example 11.11. $1 + i\sqrt{3}$: $r = 2$, $\cos\theta = \frac{1}{2}$, $\sin\theta = \frac{\sqrt{3}}{2}$, so $\theta = \frac{\pi}{3}$ and $1 + i\sqrt{3} = 2\,\mathrm{cis}\,\frac{\pi}{3}$. $-1 - i$: $r = \sqrt{2}$, third quadrant, $\theta = \frac{5\pi}{4}$, so $-1 - i = \sqrt{2}\,\mathrm{cis}\,\frac{5\pi}{4}$.

Theorem 11.12 (Polar Multiplication)**.** $r_1\,\mathrm{cis}\,\theta_1 \cdot r_2\,\mathrm{cis}\,\theta_2 = r_1 r_2\,\mathrm{cis}(\theta_1 + \theta_2)$.

Proof. Expand and use the angle-sum identities: $(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2) = (\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + i(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2) = \cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)$. □

So multiplying complex numbers multiplies moduli and *adds* arguments; multiplying by $i = \operatorname{cis}\frac{\pi}{2}$ rotates by 90°.

11.4 De Moivre's theorem

Theorem 11.13 (De Moivre's Theorem, DMT)**.** *For all $\theta \in \mathbb{R}$ and $n \in \mathbb{Z}$, $(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$.*

Proof. For $n \geq 0$, induction: $n = 0$ gives $1 = 1$, and $(\operatorname{cis}\theta)^{k+1} = \operatorname{cis}(k\theta)\operatorname{cis}\theta = \operatorname{cis}((k+1)\theta)$ by polar multiplication. For $n = -m < 0$: $(\operatorname{cis}\theta)^{-m} = 1/\operatorname{cis}(m\theta) = \operatorname{cis}(-m\theta)$, since $\operatorname{cis}(m\theta)\operatorname{cis}(-m\theta) = \operatorname{cis} 0 = 1$. □

Example 11.14. Compute $(1 + i\sqrt{3})^{10}$:

$$(2\operatorname{cis}\tfrac{\pi}{3})^{10} = 2^{10}\operatorname{cis}\tfrac{10\pi}{3} = 1024\operatorname{cis}\tfrac{4\pi}{3} = 1024\left(-\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}i\right) = -512 - 512\sqrt{3}\,i.$$

Example 11.15. $(1-i)^8 = (\sqrt{2}\operatorname{cis}(-\tfrac{\pi}{4}))^8 = 16\operatorname{cis}(-2\pi) = 16$.

Example 11.16 (Trig identities)**.** By DMT and the binomial theorem, $\cos 3\theta + i\sin 3\theta = (c+is)^3 = c^3 + 3ic^2s - 3cs^2 - is^3$ where $c = \cos\theta$, $s = \sin\theta$. Comparing real parts, $\cos 3\theta = c^3 - 3cs^2 = c^3 - 3c(1-c^2) = 4\cos^3\theta - 3\cos\theta$.

11.5 The complex exponential

Definition 11.17. For $\theta \in \mathbb{R}$ define $e^{i\theta} = \cos\theta + i\sin\theta$ (Euler's formula). For $z = x+yi$, $e^z = e^x e^{iy}$.

With this notation polar multiplication is $e^{i\theta_1}e^{i\theta_2} = e^{i(\theta_1+\theta_2)}$ and DMT reads $(e^{i\theta})^n = e^{in\theta}$, just like real exponent laws. Every non-zero z can be written $z = re^{i\theta}$. Famous special case: $e^{i\pi} + 1 = 0$. Also $\overline{e^{i\theta}} = e^{-i\theta}$, so $\cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2}$.

11.6 n-th roots

Theorem 11.18 (Complex n-th Roots Theorem, CNRT)**.** *Let $a = r\operatorname{cis}\theta \neq 0$ and $n \in \mathbb{N}$. The equation $z^n = a$ has exactly n distinct solutions*

$$z_k = \sqrt[n]{r}\,\operatorname{cis}\left(\frac{\theta + 2\pi k}{n}\right), \qquad k = 0, 1, \ldots, n-1,$$

where $\sqrt[n]{r}$ is the positive real n-th root.

Proof. Write $z = s\operatorname{cis}\varphi$. By DMT, $z^n = s^n\operatorname{cis}(n\varphi) = r\operatorname{cis}\theta$ iff $s^n = r$ and $n\varphi = \theta + 2\pi k$ for some integer k. So $s = \sqrt[n]{r}$ and $\varphi = \frac{\theta+2\pi k}{n}$. Values of k that are congruent mod n give the same angle up to a multiple of 2π, and $k = 0, \ldots, n-1$ give n angles in $[\frac{\theta}{n}, \frac{\theta}{n} + 2\pi)$, which are distinct points. □

The n-th roots lie on a circle of radius $\sqrt[n]{r}$, equally spaced at angles $2\pi/n$: the vertices of a regular n-gon.

Example 11.19. Solve $z^3 = 8i$. $8i = 8\operatorname{cis}\frac{\pi}{2}$, so $z_k = 2\operatorname{cis}\left(\frac{\pi}{6} + \frac{2\pi k}{3}\right)$:

$$\begin{aligned} z_0 &= 2\operatorname{cis}\tfrac{\pi}{6} = \sqrt{3} + i, \\ z_1 &= 2\operatorname{cis}\tfrac{5\pi}{6} = -\sqrt{3} + i, \\ z_2 &= 2\operatorname{cis}\tfrac{3\pi}{2} = -2i. \end{aligned}$$

Check: $(-2i)^3 = -8i^3 = 8i$.

Example 11.20. Solve $z^4 = -16$. $-16 = 16\operatorname{cis}\pi$, so $z_k = 2\operatorname{cis}\frac{\pi+2\pi k}{4}$, $k = 0, 1, 2, 3$: $\pm\sqrt{2} \pm \sqrt{2}i$ (all four sign combinations).

11.7 Roots of unity

Definition 11.21. The solutions of $z^n = 1$ are the *n-th roots of unity*: $\omega_k = \operatorname{cis}\frac{2\pi k}{n} = e^{2\pi ik/n}$, $k = 0, \ldots, n-1$. With $\omega = \omega_1$, they are $1, \omega, \omega^2, \ldots, \omega^{n-1}$.

Example 11.22. Cube roots of unity: 1, $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$, $\omega^2 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$. Since $z^3 - 1 = (z-1)(z^2 + z + 1)$, we get $1 + \omega + \omega^2 = 0$. Sixth roots of unity: ± 1, $\pm\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$.

Proposition 11.23. *For $n \geq 2$, the n-th roots of unity sum to 0: $\sum_{k=0}^{n-1} \omega^k = \frac{\omega^n - 1}{\omega - 1} = 0$. If $z^n = a$ has a solution z_0, then all solutions are $z_0\omega^k$.*

11.8 Common pitfalls

- Calling bi the imaginary part; $\operatorname{Im}(a + bi) = b$, a real number.
- Choosing the wrong quadrant for the argument from $\arctan(y/x)$.
- Giving only one n-th root, or listing $n + 1$ of them (the $k = n$ root repeats $k = 0$).
- Writing $\sqrt{z}$ for complex z as though it were a single well-defined value.
- $|z + w| \neq |z| + |w|$ in general.

11.9 Summary

- $\mathbb{C}$ is a field; divide by multiplying by the conjugate; $z\overline{z} = |z|^2$.
- Polar form $z = r\operatorname{cis}\theta = re^{i\theta}$; multiplication multiplies moduli and adds arguments.
- DMT: $(\operatorname{cis}\theta)^n = \operatorname{cis}(n\theta)$, for computing powers and deriving identities.
- CNRT: $z^n = r\operatorname{cis}\theta$ has n roots $\sqrt[n]{r}\operatorname{cis}\frac{\theta+2\pi k}{n}$; the n-th roots of unity are $e^{2\pi ik/n}$ and sum to 0.

Chapter 12

Polynomials

12.1 Definitions

Definition 12.1. Let $\mathbb{F}$ be one of $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ or $\mathbb{Z}_p$ (p prime). A *polynomial* over $\mathbb{F}$ is an expression $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ with coefficients $a_i \in \mathbb{F}$. If $a_n \neq 0$, the *degree* is $\deg f = n$ and a_n is the *leading coefficient*; f is *monic* if $a_n = 1$. The zero polynomial has no degree. The set of all such polynomials is $\mathbb{F}[x]$.

Polynomials are added and multiplied in the usual way. Over a field, $\deg(fg) = \deg f + \deg g$ for non-zero f, g (Degree of a Product). Two polynomials are equal iff all corresponding coefficients are equal.

Definition 12.2. $c \in \mathbb{F}$ is a *root* (zero) of f if $f(c) = 0$. g *divides* f, $g \mid f$, if $f = gh$ for some $h \in \mathbb{F}[x]$.

12.2 Division algorithm for polynomials

Theorem 12.3 (Division Algorithm for Polynomials, DAP)**.** *Let $f, g \in \mathbb{F}[x]$ with $g \neq 0$. There exist unique $q, r \in \mathbb{F}[x]$ with*

$$f = qg + r, \qquad r = 0 \text{ or } \deg r < \deg g.$$

Proof sketch. Existence: long division, i.e. repeatedly subtract a multiple $\frac{\text{lead}(f)}{\text{lead}(g)} x^{\deg f - \deg g} g$ of g to kill the leading term; the degree drops each time (formally, induction on $\deg f$). Uniqueness: if $q_1 g + r_1 = q_2 g + r_2$ then $(q_1 - q_2)g = r_2 - r_1$; if $q_1 \neq q_2$ the left side has degree $\geq \deg g$ while the right has degree $< \deg g$, impossible. □

Example 12.4. Divide $f(x) = 2x^4 + 3x^3 - x + 5$ by $g(x) = x^2 + x - 1$.

- $2x^4/x^2 = 2x^2$; subtract $2x^2 g = 2x^4 + 2x^3 - 2x^2$: remainder so far $x^3 + 2x^2 - x + 5$.
- $x^3/x^2 = x$; subtract $xg = x^3 + x^2 - x$: get $x^2 + 5$.
- $x^2/x^2 = 1$; subtract $g = x^2 + x - 1$: get $-x + 6$.

So $q(x) = 2x^2 + x + 1$ and $r(x) = -x + 6$. Check: $(x^2 + x - 1)(2x^2 + x + 1) = 2x^4 + 3x^3 - 1$, and adding $-x + 6$ gives f.

12.3 Remainder and factor theorems

Theorem 12.5 (Remainder Theorem, RT)**.** *The remainder when $f(x)$ is divided by $x - c$ is $f(c)$.*

Proof. By DAP, $f(x) = q(x)(x - c) + r$ with r a constant (degree < 1). Substitute $x = c$: $f(c) = r$. □

Theorem 12.6 (Factor Theorem, FT)**.** *$(x - c) \mid f(x)$ if and only if $f(c) = 0$.*

Example 12.7. The remainder of $f(x) = x^5 + 2x^4 - x^3 + 3x - 7$ upon division by $x + 2$ is $f(-2) = -32 + 32 + 8 - 6 - 7 = -5$.

Corollary 12.8. *A non-zero polynomial of degree n over a field has at most n roots.*

Proof. Induction on n: if c is a root, $f = (x - c)q$ with $\deg q = n - 1$; any other root $d \neq c$ satisfies $(d - c)q(d) = 0$, so $q(d) = 0$, and q has at most $n - 1$ roots. □

This uses that $\mathbb{F}$ has no zero divisors. In $\mathbb{Z}_8$, $x^2 - 1$ has four roots $1, 3, 5, 7$. Compare the four square roots of 4 mod 35 in Chapter 9.

12.4 The Fundamental Theorem of Algebra

Theorem 12.9 (Fundamental Theorem of Algebra, FTA)**.** *Every non-constant polynomial in $\mathbb{C}[x]$ has at least one root in $\mathbb{C}$.*

(The proof requires analysis and is not done in Math 135.) Combined with the factor theorem and induction on the degree:

Corollary 12.10 (Complex Polynomials of Degree n Have n Roots, CPN)**.** *Every $f \in \mathbb{C}[x]$ of degree $n \geq 1$ factors as*

$$f(x) = a_n(x - c_1)(x - c_2) \cdots (x - c_n)$$

with $c_1, \ldots, c_n \in \mathbb{C}$ (not necessarily distinct). So f has exactly n roots counted with multiplicity.

The *multiplicity* of a root c is the largest k with $(x - c)^k \mid f$.

Example 12.11. $z^6 - 1 = \prod_{k=0}^{5}(z - \omega^k)$ where $\omega = e^{i\pi/3}$; the roots are the sixth roots of unity.

12.5 Real polynomials: conjugate roots and real factorization

Theorem 12.12 (Conjugate Roots Theorem, CJRT)**.** *If $f \in \mathbb{R}[x]$ and $c \in \mathbb{C}$ is a root of f, then $\overline{c}$ is also a root.*

Proof. Since the coefficients are real, $\overline{a_k} = a_k$, so $f(\overline{c}) = \sum a_k \overline{c}^k = \overline{\sum a_k c^k} = \overline{f(c)} = \overline{0} = 0$ (PCJ). □

For a non-real root c, the two factors $x - c$ and $x - \overline{c}$ combine into a *real* quadratic:

$$(x - c)(x - \overline{c}) = x^2 - 2\,\mathrm{Re}(c)\,x + |c|^2\,.$$

Theorem 12.13 (Real Factorization)**.** *Every $f \in \mathbb{R}[x]$ of degree ≥ 1 can be written as a product of real linear factors and real quadratic factors with negative discriminant (irreducible over $\mathbb{R}$). In particular, a real polynomial of odd degree has a real root.*

Example 12.14. Given that $2i$ is a root of $f(x) = x^4 - 2x^3 + 6x^2 - 8x + 8$, factor f over $\mathbb{R}$ and $\mathbb{C}$.

By CJRT, $-2i$ is also a root, so $(x - 2i)(x + 2i) = x^2 + 4$ divides f. Long division: $f(x) = (x^2 + 4)(x^2 - 2x + 2)$; check: $x^4 - 2x^3 + 2x^2 + 4x^2 - 8x + 8 = f(x)$. The quadratic $x^2 - 2x + 2$ has discriminant $4 - 8 = -4 < 0$, roots $1 \pm i$. Hence

$$\begin{aligned} &\text{over } \mathbb{R}: \quad f(x) = (x^2 + 4)(x^2 - 2x + 2), \\ &\text{over } \mathbb{C}: \quad f(x) = (x - 2i)(x + 2i)(x - 1 - i)(x - 1 + i). \end{aligned}$$

Example 12.15. Factor $x^6 - 1$ over $\mathbb{R}$. Its roots are the sixth roots of unity ± 1, $\frac{1}{2} \pm \frac{\sqrt{3}}{2} i$, $-\frac{1}{2} \pm \frac{\sqrt{3}}{2} i$. Pairing conjugates ($\mathrm{Re} = \pm\frac{1}{2}$, $|c|^2 = 1$):

$$x^6 - 1 = (x - 1)(x + 1)(x^2 - x + 1)(x^2 + x + 1).$$

CJRT needs real coefficients

$f(z) = z - i$ has root i but not $-i$. The theorem fails for polynomials with non-real coefficients.

12.6 Rational roots

Theorem 12.16 (Rational Roots Theorem, RRT)**.** *Let $f(x) = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ with $a_n \neq 0$. If $\frac{p}{q}$ is a root in lowest terms ($\gcd(p, q) = 1$, $q > 0$), then $p \mid a_0$ and $q \mid a_n$.*

Proof. Multiply $f(p/q) = 0$ by q^n: $a_n p^n + a_{n-1} p^{n-1} q + \cdots + a_1 p q^{n-1} + a_0 q^n = 0$. Every term except $a_0 q^n$ is a multiple of p, so $p \mid a_0 q^n$; as $\gcd(p, q^n) = 1$, CAD gives $p \mid a_0$. Similarly every term except $a_n p^n$ is a multiple of q, so $q \mid a_n$. □

Example 12.17. Factor $f(x) = 2x^3 + x^2 - 7x - 6$ over $\mathbb{Q}$. Candidates p/q with $p \mid 6$, $q \mid 2$: $\pm 1, \pm 2, \pm 3, \pm 6, \pm\frac{1}{2}, \pm\frac{3}{2}$. Try $x = -1$: $-2 + 1 + 7 - 6 = 0$. So $(x + 1) \mid f$, and division gives $f(x) = (x + 1)(2x^2 - x - 6)$. Factoring the quadratic, $2x^2 - x - 6 = (2x + 3)(x - 2)$, so

$$f(x) = (x + 1)(2x + 3)(x - 2),$$

with roots $-1, -\frac{3}{2}, 2$, all among the candidates.

Example 12.18. $x^3 - 3x - 1$ has no rational roots: the only candidates are ± 1, and $f(1) = -3$, $f(-1) = 1$. Since a cubic that factors over $\mathbb{Q}$ must have a linear factor, $x^3 - 3x - 1$ is irreducible over $\mathbb{Q}$. (It still has three real roots, by the intermediate value theorem.)

12.7 Polynomials over $\mathbb{Z}_p$

The division algorithm and factor theorem work over $\mathbb{Z}_p$ as well. For example, over $\mathbb{Z}_5$, $x^2 + 1$ has roots 2 and 3 ($4 + 1 = 5 \equiv 0$, $9 + 1 = 10 \equiv 0$), so $x^2 + 1 = (x - 2)(x - 3) = (x + 3)(x + 2)$ in $\mathbb{Z}_5[x]$, while over $\mathbb{Z}_3$ it has no roots ($0^2 + 1 = 1$, $1^2 + 1 = 2$, $2^2 + 1 = 5 \equiv 2$) and is irreducible. By FLT, $x^p - x = \prod_{a \in \mathbb{Z}_p} (x - a)$ in $\mathbb{Z}_p[x]$.

12.8 Common pitfalls

- Forgetting zero coefficients when doing long division (write $2x^4 + 3x^3 + 0x^2 - x + 5$).
- Applying CJRT to a polynomial with complex coefficients.
- Applying RRT to non-integer coefficients (clear denominators first) or forgetting negative candidates.
- Stopping at "irreducible over $\mathbb{R}$" when the question asks for a factorization over $\mathbb{C}$.

12.9 Summary

- DAP: $f = qg + r$ with $\deg r < \deg g$. RT: $f(c)$ is the remainder mod $x - c$; FT: $f(c) = 0 \iff (x - c) \mid f$.
- FTA: every non-constant complex polynomial has a root; degree n means exactly n roots with multiplicity.
- CJRT: real polynomials have non-real roots in conjugate pairs; real polynomials factor into real linear and irreducible quadratic factors.
- RRT: rational roots p/q of integer polynomials have $p \mid a_0$, $q \mid a_n$.